A Project Report on

PADDY CROP DISEASE RECOGNITION WITH MACHINE LEARNING TECHNIQUES

Submitted to

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In the partial fulfillment for the award of the degree of

POST GRADUATION

IN

MASTER OF COMPUTER APPLICATIONS

By

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UNDER THE GUIDANCE OF

B. MURALI KRISHNA,MCA

RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS

(AUTONOMOUS)

(Approved by AICTE-NEWDELHI,Affiliated to JNTUK,KAKINADA)

(AnISO9001:2015 certified Institute,Accredited by NAAC with 'A' Grade&NBA)

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(2023-2025)

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BONAFIDECERTIFICATE

This is to certify that this project work entitled **"PADDY CROP DISEASE RECOGNITION WITH MACHINE LEARNING TECHNIQUES"** is carried out by **Ms.KANUGULA JAHNAVI (238A1F0015)** in partial fulfillment of the requirement for the Award of Degree in **MASTER OF COMPUTER APPLICATIONS,** for the academic year **2024 – 2025.** This work was done under my supervision and guidance.

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STUDENT DECLARATION

I hereby declare that this report for the Project Title "**PADDY CROP DISEASE RECOGNITION WITH MACHINE LEARNING TECHNIQUES"** has been developed by me under the able supervision of Assistant Professor **Mr.B.MURALI KRISHNA, MCA** and submitted to Department of Master of Computer Applications, **RISE Krishna Sai Prakasam Group of Institutions,Autonomous,Valluru,Ongole** for partial fulfillment of my project work.It is my own project at this college/university and published by me.

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ACKNOWLEDGEMENT

I express my sincere thanks to the Chairman **Sri I.C. RANGAMANNAR**, Vice Chairman **Sri S. VENKATESWARA RAO**, Secretary and Correspondent **Sri S. HANUMANTHARAO**, and Treasurer **Sri S.BHARATH,** and **RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS**(AUTONOMOUS)who have provided all necessary facilities to carry out the project effectively.

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ABSTRACT

Nowadays, paddy is a very important crop grown in many countries, especially in India. It is the main food for many people and is grown on a large scale. However, paddy plants often get affected by different types of diseases. These diseases reduce the amount and quality of rice produced. Because of this, farmers face big losses every year. Mostly, farmers check the plants by looking at the leaves with their eyes. This method is slow and sometimes not correct. If farmers do not find the disease early, the crops may get fully damaged. So, it is very important to detect diseases quickly and correctly. This project helps to find paddy leaf diseases automatically. We use image processing to read and understand the leaf images. Images of healthy and diseased leaves are collected and cleaned. Important details from the leaves are taken out to help the system. We use a method called CNN (Convolutional Neural Network) to find the disease. CNN helps the system to check and give the correct disease name. This system is fast and gives better results than manual checking. It helps farmers to find the disease in the early stage. Farmers can take action quickly to save the crops. This system helps improve rice production and reduce the loss. The main aim is to make farmers' work easy and support better farming.

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1. INTRODUCTION

Rice, also known as paddy, is a very important food crop that is grown in many countries. However, paddy plants often get infected by diseases caused by bacteria, fungi, or viruses. These diseases can spread quickly and reduce the crop yield, leading to financial losses for farmers. In the past, farmers would check the plants with their own eyes to identify diseases, but this method is slow and not always correct. Now, with the help of machine learning, farmers can detect diseases faster and more accurately using computer programs.

The first step in using machine learning for disease detection is collecting images of healthy and diseased paddy leaves. These images come from farms, agricultural research centers, or online sources. Once collected, the images go through processing to improve their quality. The computer adjusts brightness, removes unnecessary parts, and focuses on the affected areas of the leaves to make the disease signs clearer.

Next, the system looks for important features in the leaves, such as changes in color, texture, and shape. Each disease has unique symptoms—for example, bacterial leaf blight causes yellow streaks on leaves, while blast disease creates dark spots. The computer studies these differences to recognize the disease type.

To train the machine learning model, different techniques are used. Some common methods include Support Vector Machines (SVM) and Decision Trees, but the best results come from Convolutional Neural Networks (CNNs). CNNs are a type of artificial intelligence that can analyze thousands of images and learn patterns, helping the model detect diseases with high accuracy. Once trained, the model can quickly examine a new image of a paddy leaf and tell whether it is healthy or affected by a disease.

To overcome these limitations, researchers have developed automated disease detection systems by combining modern technologies such as image processing, machine learning (ML), deep learning (DL), and computer vision. In these systems, images of healthy and diseased rice plants, especially leaves, are captured under different conditions. These images undergo pre-processing techniques like filtering, noise removal, and resizing to improve image quality. Important features such as color, texture, shape, and spot patterns are extracted from the pre-processed images, which serve as key indicators for disease identification. Once the features are extracted, classification models like Support Vector Machine (SVM), Random Forest, or advanced deep learning models like Convolutional Neural Networks (CNN) are employed to accurately classify and detect the specific disease affecting the crop.

The implementation of automated disease detection systems brings numerous advantages to modern agriculture. These systems provide farmers with faster, more reliable, and objective results compared to traditional manual methods. By enabling early detection of diseases, farmers can take timely actions to control the spread, apply proper treatments, and prevent major yield loss. Additionally, it promotes sustainable farming by minimizing the excessive use of chemical pesticides, thus reducing environmental pollution. The use of AI-based disease identification also reduces the dependency on human experts, making it more accessible to farmers, especially in remote areas.

With the advancement of technology, these automated systems are now available in the form of mobile applications, drone-based monitoring tools, and smart farming devices, making them easy to use even for small-scale farmers. This integration of machine learning and agriculture is leading towards the concept of smart and precision farming, where crops are continuously monitored, and necessary actions are taken automatically. Ultimately, paddy crop disease identification using AI not only helps in reducing economic losses but also ensures better food security and promotes environmentally friendly agricultural practices.

To make sure the model works well, it is tested for accuracy before being used in real life. A well-trained model can detect diseases quickly and correctly, helping farmers take early action to save their crops.

There are many benefits to using machine learning for disease detection. Farmers can find diseases earlier and treat them before they spread. This method is also faster than checking plants manually. Another advantage is that it helps reduce the use of pesticides, as farmers can treat only the affected areas instead of spraying the entire field. This saves money, time, and protects the environment.

In short, machine learning makes it easier for farmers to detect paddy diseases**,** protect their crops, and increase their yield. This technology is improving farming by making it smarter, faster, and more efficient.

2.LITERATURE SURVEY

TITLE: Paddy Crop Disease Detection Using Machine Learning

Authors: S. Girinath and Hemanth Chalamkota

This research focuses on developing a machine learning model to detect and classify paddy crop diseases, utilizing image processing techniques to analyze leaf patterns and symptoms.

TITLE: Paddy Leaf Diseases Identification on Infrared Images Based on Convolutional Neural Networks

Authors: Petchiammal A, Briskline Kiruba S, and D. Murugan

The paper proposes a CNN-based model for identifying paddy leaf diseases using infrared imagery, achieving significant accuracy in classifying multiple disease types.

TITLE: Paddy Doctor: A Visual Image Dataset for Automated Paddy Disease Classification and Benchmarking

Authors: Petchiammal A, Briskline Kiruba S, D. Murugan, and Pandarasamy A

This study introduces the 'Paddy Doctor' dataset, comprising annotated images of paddy leaves affected by various diseases, and benchmarks several deep learning models for automated disease classification.

TITLE: Deep Learning Based Models for Paddy Disease Identification and Classification: A Systematic Survey

Authors: S. K. Dubey, S. K. Maurya, and S. K. Singh

This survey paper provides a comprehensive overview of deep learning models applied to paddy disease identification and classification, discussing various architecture and their performance me

TITLE: Detection of Paddy Diseases using Deep Learning Methodologies

Authors: S. K. Dubey, S. K. Maurya, and S. K. Singh

The study explores deep learning methodologies for detecting paddy diseases, focusing on the implementation of neural networks to improve diagnostic accuracy.

TITLE: AgriCare: To Detect and Prevent Paddy Crop Diseases Using Machine Learning

Authors: S. K. Dubey, S. K. Maurya, and S. K. Singh

This paper proposes the 'AgriCare' system, which utilizes machine learning techniques to identify and prevent diseases in paddy crops, aiming to support farmers in maintaining crop health.

TITLE: Prediction of Paddy Crop Diseases Using Convolutional Neural Networks

Authors: S. K. Dubey, S. K. Maurya, and S. K. Singh

The research focuses on predicting paddy crop diseases by employing Convolutional Neural Networks, highlighting the model's effectiveness in early disease detection and prevention strategies.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing paddy crop disease detection system heavily relies on farmers' visual inspection and traditional knowledge, which can be subjective and inconsistent. These methods often lead to delayed diagnosis and improper treatment of diseases. In many cases, diseases are identified at advanced stages, reducing the effectiveness of interventions. While some mobile applications provide basic disease information, they lack the capability for accurate, real-time detection based on actual crop images. Additionally, farmers in rural or remote areas have limited access to expert advice, making it difficult to manage crop health effectively. This calls for a more advanced, automated solution to provide timely and accurate disease detection.

3.1.1 DISADVANTAGES

- The existing system is error-prone due to manual observation.

3.2 PROPOSED SYSTEM

The proposed system uses machine learning to detect paddy crop diseases from images. It analyzes high-resolution images along with environmental data such as temperature and humidity. The system provides real-time diagnosis and classifies diseases based on symptoms. It also assesses the severity of the disease for timely intervention. Farmers receive actionable treatment recommendations through the system. By automating the process, the system improves accuracy and efficiency in disease detection.

3.2.1 ADVANTAGES

- The proposed system offers real-time, accurate disease detection.

3.3 SYSTEM REQUIREMENTS

3.3.1 HARDWARE REQUIREMENTS

- System : Pentium IV

- RAM :6GB(min)

- Hard Disk : 500 GB
- Input devices: Keyboard, Mouse
- Monitor : 15”LED

3.3.2 SOFTWARE REQUIREMENTS

- Operating System : Windows10
- Front End : HTML, Js,css
- Web Framework : Django
- Coding Language :python
- Databas : MySQL 5.0

3.4 SYSTEM STUDY

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ♦ ECONOMICAL FEASIBILITY
- ♦ TECHNICAL FEASIBILITY
- ♦ SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and

development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

4. SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

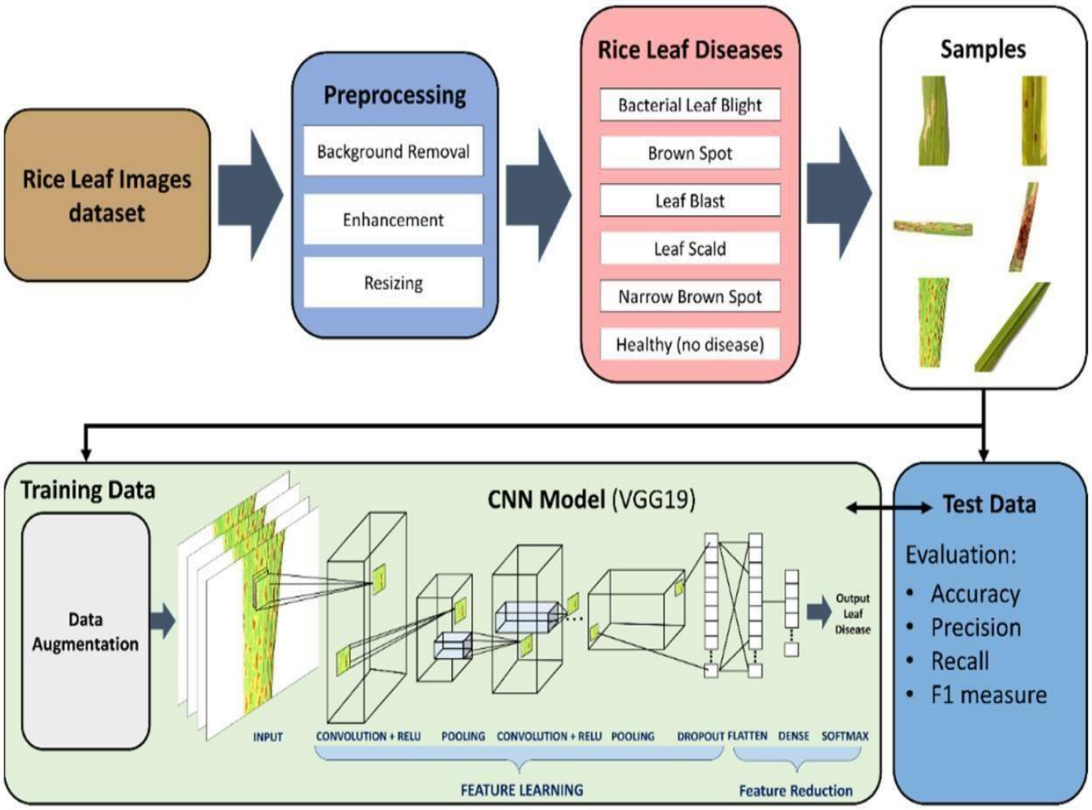


Fig4.1:SYSTEM ARCHITECTURE

The presented architecture is designed to automatically detect and classify diseases in rice leaves using deep learning techniques, specifically the VGG19 Convolutional Neural Network (CNN) model. The process starts with the collection of a rice leaf image dataset that includes various images of diseased and healthy paddy leaves. These images are collected from fields, research centers, or online databases. The next step is preprocessing, where the images are prepared for analysis by performing background removal, image enhancement, and resizing. This step is crucial to eliminate unnecessary background noise and adjust the image to the desired input size of the CNN model.

After preprocessing, the system classifies the leaves into six classes: Bacterial Leaf Blight, Brown Spot, Leaf Blast, Leaf Scald, Narrow Brown Spot, and Healthy (no

disease). The samples of these categories are shown to give a visual representation of how infected leaves appear. Once categorized, the dataset is divided into training and testing sets. Before training, data augmentation techniques such as image rotation, flipping, and zooming are applied to the training data to increase the diversity of the dataset and improve the model's generalization ability.

The core of the system is the VGG19 CNN model, which is used for feature extraction and disease classification. This model consists of several convolutional layers followed by ReLU activation, pooling layers for dimensionality reduction, fully connected dense layers, and a softmax layer for final classification. The model automatically learns the important features from the images like color variations, texture patterns, and leaf spot shapes which are essential for detecting diseases. Finally, the trained model is tested using the test dataset to evaluate its performance. The evaluation is done using metrics such as accuracy, precision, recall, and F1-score, which determine the reliability and effectiveness of the model in correctly identifying paddy leaf diseases.

This automated system helps in accurate and fast identification of paddy diseases without the need for manual observation, enabling farmers to take preventive and corrective measures quickly, thus improving crop productivity and minimizing yield losses.

4.2 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven

successful in the modeling of large and complex systems.

The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS

The Primary goals in the design of the UML are as follows:

Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.

Provide extendibility and specialization mechanisms to extend the core concepts.

Be independent of particular programming languages and development process.

Provide a formal basis for understanding the modeling language.

Encourage the growth of OO tools market.

Support higher level development concepts such as collaborations, frameworks, patterns and components.

Integrate best practices.

4.3 USE CASE DIAGRAM

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

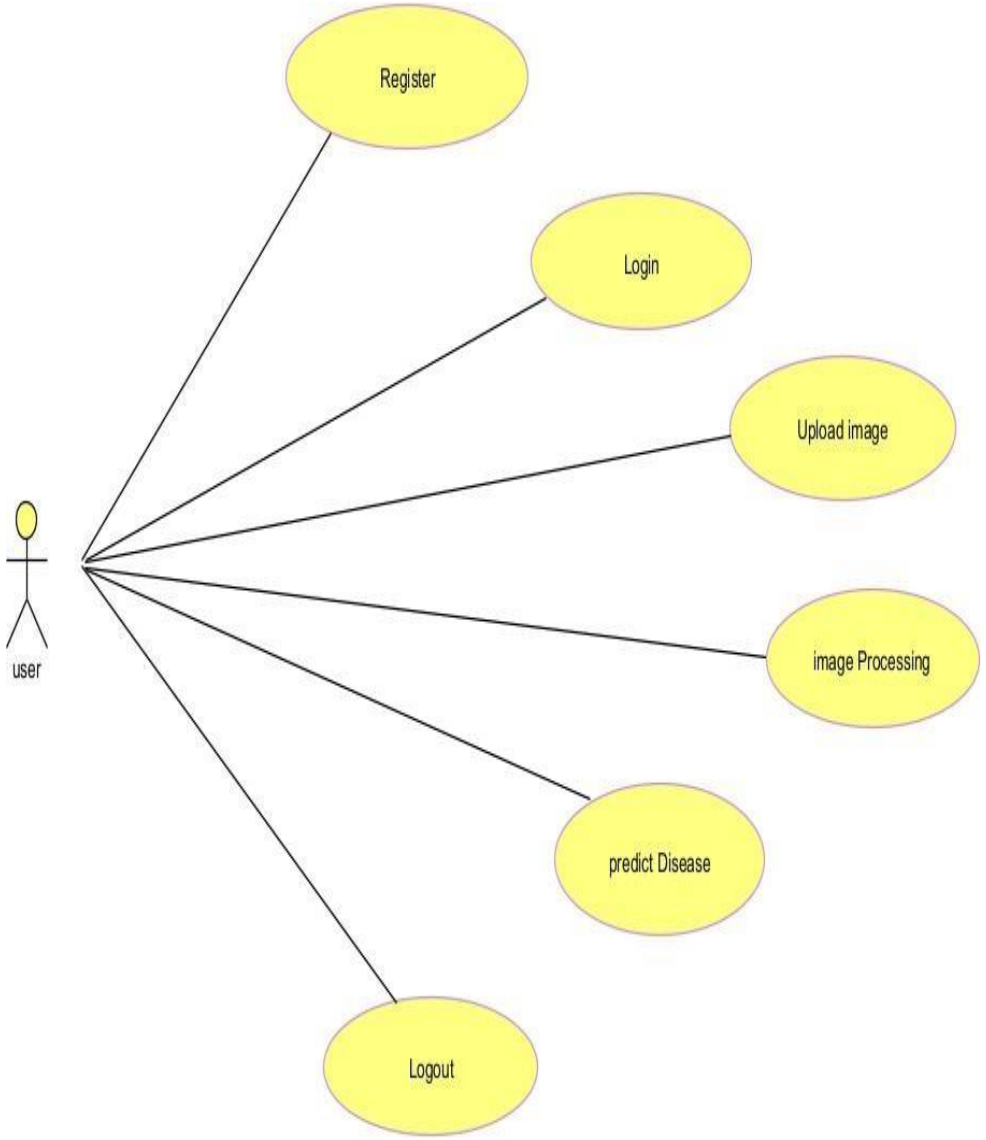


Fig 4.2 USECASE DIAGRAM

4.4 CLASS DIAGRAM

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information

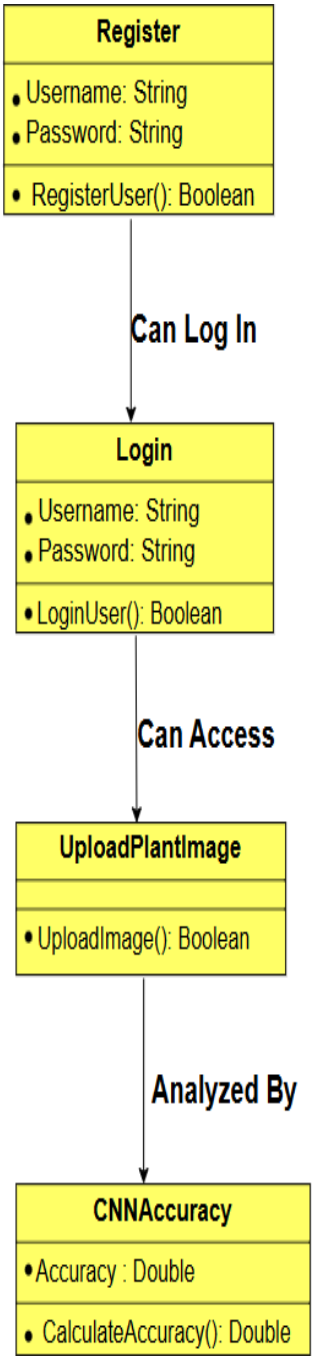


Fig 4.3 CLASS DIAGRAM

4.5 SEQUENCE DIAGRAM

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams

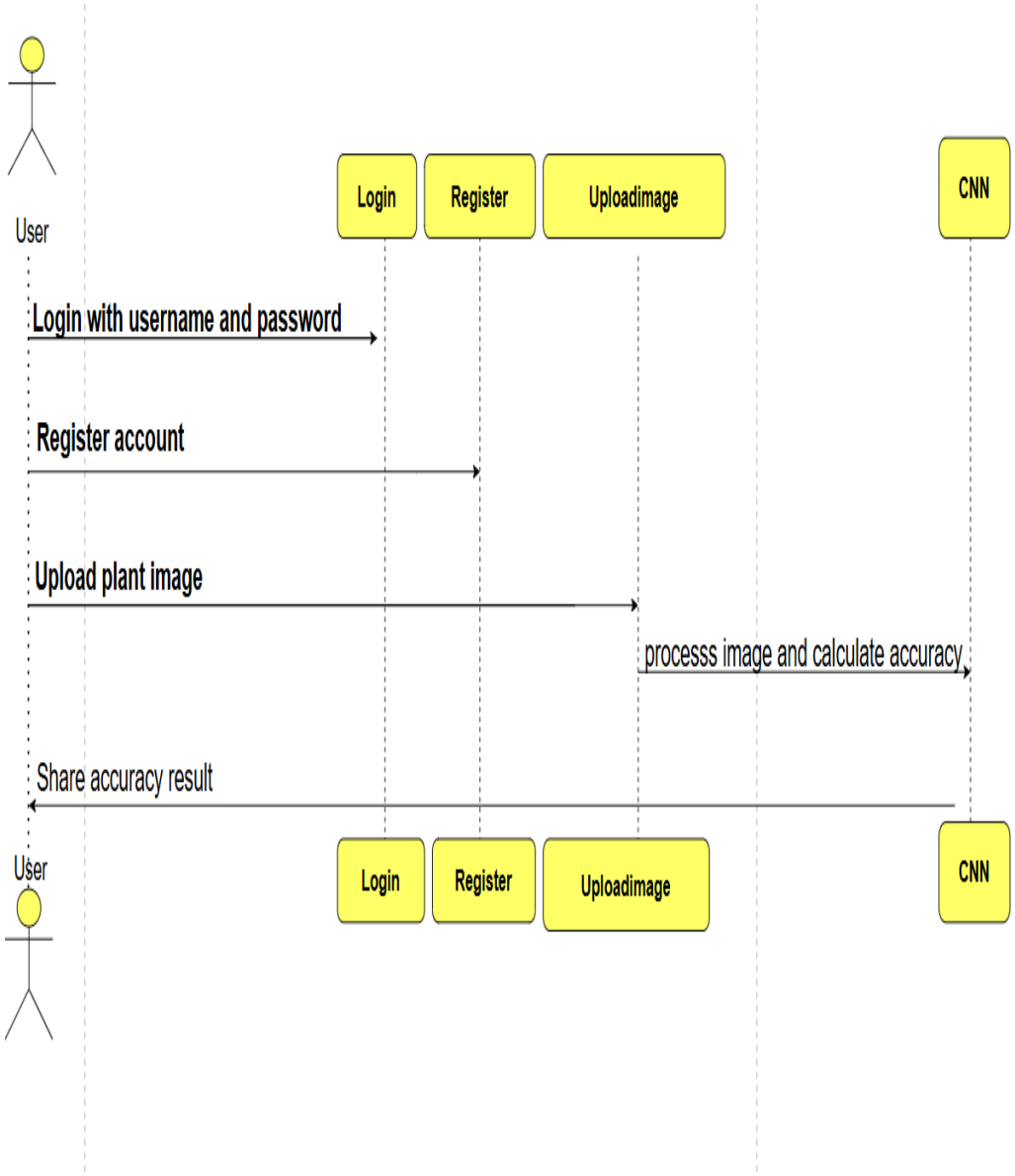


Fig4.4 SEQUENCE DIAGRAM

4.6 INPUT &OUTPUT DESIGNS

4.6.1 INPUT DESIGNS

The input design for a paddy crop disease detection system focuses on collecting high-quality data to ensure accurate disease identification. The primary input consists of images of paddy crops, captured using mobile phones, drones, or cameras. These images must be of high resolution to allow the machine learning model to detect disease symptoms accurately. Additionally, environmental parameters such as temperature, humidity, and soil moisture levels, collected through IoT-based sensors, serve as supplementary inputs to enhance prediction accuracy. Farmers or agricultural experts may also provide manual inputs, such as specific observations of symptoms like leaf discoloration, spots, or wilting.

To maintain the quality of input data, various validation techniques are applied. Image quality checks ensure that only clear and focused images are processed, while mandatory fields for environmental data and metadata such as location, date, and time improve reliability. Preprocessing methods such as noise reduction, contrast enhancement, and segmentation help refine images before analysis. The input interface should be user-friendly, allowing seamless data submission. Additionally, real-time feedback mechanisms should alert users if the images are blurry or do not meet the required criteria, prompting them to upload better-quality images for more accurate disease detection results.

4.6.2 OUTPUT DESIGN

The output design of the paddy crop disease detection system ensures that the processed information is presented in a clear, actionable, and user-friendly manner. Once the system analyzes the input images and environmental data, it provides disease classification results, indicating whether the crop is affected by conditions such as Bacterial Blight, Brown Spot, or Leaf Blast. The system also assesses the severity of the disease and offers confidence scores to indicate the reliability of the prediction. To enhance usability, segmented images can highlight the affected areas of the plant, and graphical representations can show disease trends over time.

The format of output delivery is designed to be accessible and convenient for farmers and agricultural experts. The system can provide real-time notifications via mobile apps, SMS alerts, or downloadable PDF reports that contain disease details, symptoms, causes, and recommended treatments. To ensure clarity, the output should be structured with simple language, color-coded indicators, and interactive elements for deeper analysis. If the system cannot confidently classify a disease, it should suggest consulting an expert or provide links to agricultural resources. Additionally, error messages should notify users if the input data is insufficient or of poor quality, guiding them to take corrective actions. A well-designed output system helps farmers make informed decisions, ensuring timely intervention to protect their crops and improve yield.

5. SYSTEM IMPLEMENTATION

5.1 MODULES

My project contains 2 modules

1. Login
2. Register

5.1.1 Login module:

The login module allows registered users (such as farmers, researchers, or administrators) to securely access the system. Users need to enter their credentials, such as a username/email and password, to authenticate their identity. The system verifies these credentials against stored data in the database. If authentication is successful, the user is granted access to the system's features, such as uploading images for disease detection and viewing recommendations. Security measures like password encryption and multi-factor authentication can be implemented to enhance data protection.

5.1.2 Register module

The registration module enables new users to create an account on the system. Users must provide essential details like name, email, contact number, and password. Once submitted, the system stores this information in a database, and in some cases, a verification process (such as email or OTP verification) is required to confirm the registration. After successful registration, users can log in and access system functionalities. This module ensures that only authenticated users interact with the system, preventing unauthorized access.

5.2 SOFTWARE ENVIRONMENT

Machine Learning Technology:

Machine learning is a discipline that deals with programming the systems so as to make them automatically learn and improve with experience. Here, learning implies recognizing and understanding the input data and taking informed decisions based on the supplied data. It is very difficult to consider all the decisions based on all possible

inputs. To solve this problem, algorithms are developed that build knowledge from a specific data and past experience by applying the principles of statistical science, probability, logic, mathematical optimization, reinforcement learning, and control theory.

Steps Involved in Machine Learning:

A machine learning project involves the following steps

- Defining a Problem
- Preparing Data
- Evaluating Algorithms
- Improving Results
- Presenting Results

Machine Learning (ML) is an automated learning with little or no human intervention. It involves programming computers so that they learn from the available inputs. The main purpose of machine learning is to explore and construct algorithms that can learn from the previous data and make predictions on new input data.

The input to a learning algorithm is training data, representing experience, and the output is any expertise, which usually takes the form of another algorithm that can perform a task. The input data to a machine learning system can be numerical, textual, audio, visual, or multimedia.

Machine Learning vs. Traditional Programming:

Traditional programming differs significantly from machine learning. In traditional programming, programmers code all the rules in consultation with an expert in the industry for which software is being developed. Each rule is based on a logical

foundation; the machine will execute an output following the logical statement. When the system grows complex, more rules need to be written. It can quickly become sustainable to maintain.

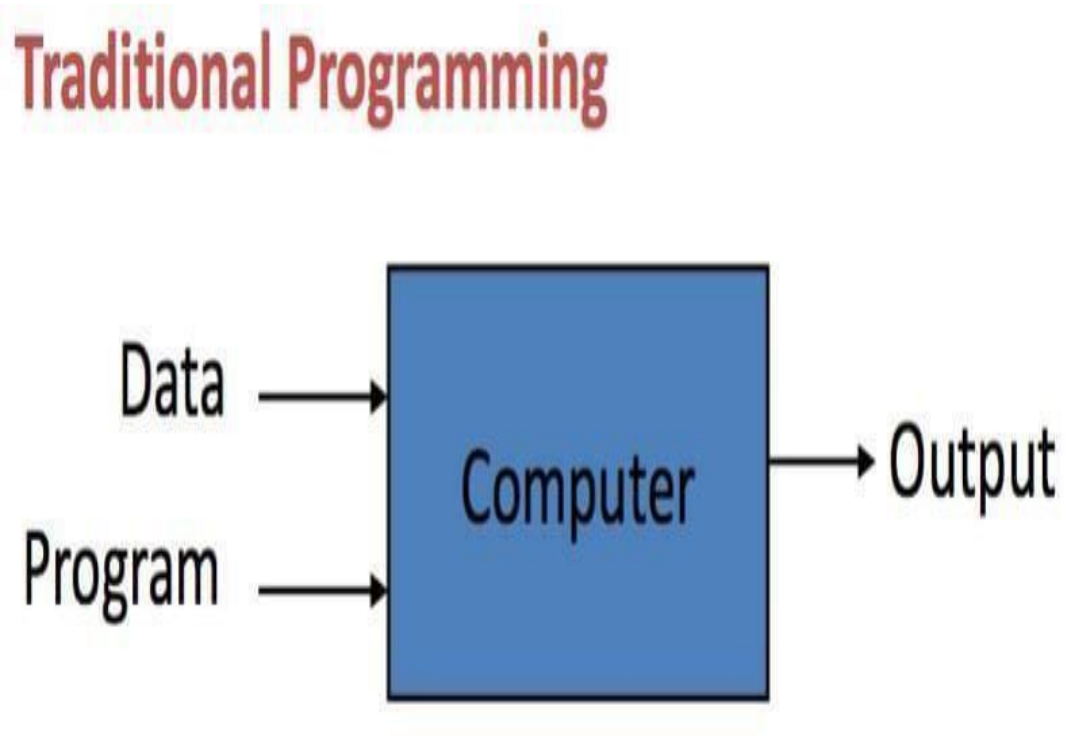


Fig 5.1: Traditional Programming

Machine learning is supposed to overcome this issue. The machine learns how the input and output data are correlated and it writes a rule. The programmers do not need to write new rules each time there is new data. The algorithms adapt in response to new data and experiences to improve efficacy over time.

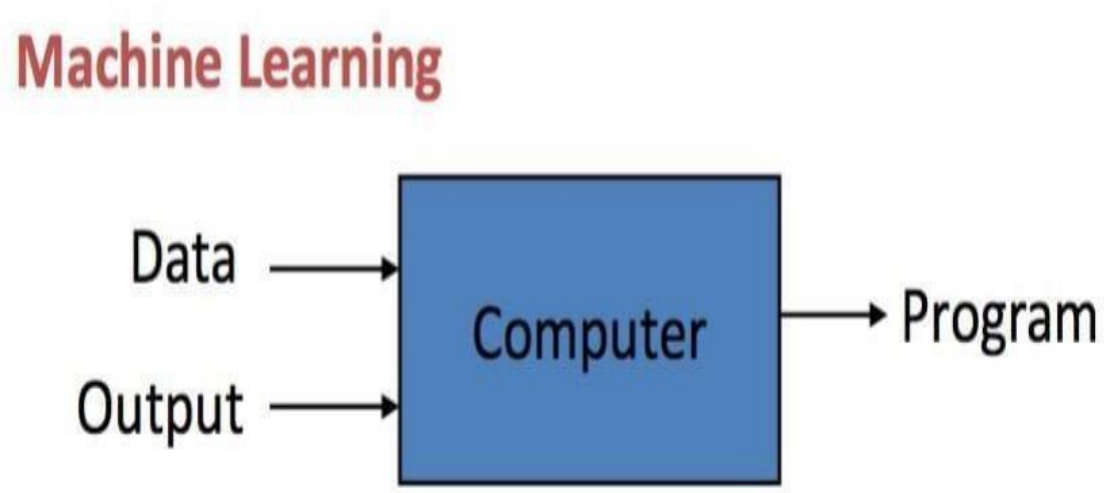


Fig5.2: Machine Learning

How does Machine learning work?

Machine learning is the brain where all the learning takes place. The way the machine learns is similar to the human being. Humans learn from experience. The more we know, the more easily we can predict. By analogy, when we face an unknown situation, the likelihood of success is lower than the known situation. Machines are trained the same. To make an accurate prediction, the machine sees an example. When we give the machine a similar example, it can figure out the outcome. However, like a human, if it's feed a previously unseen example, the machine has difficulties to predict.

Machine learning Algorithms and where they are used?

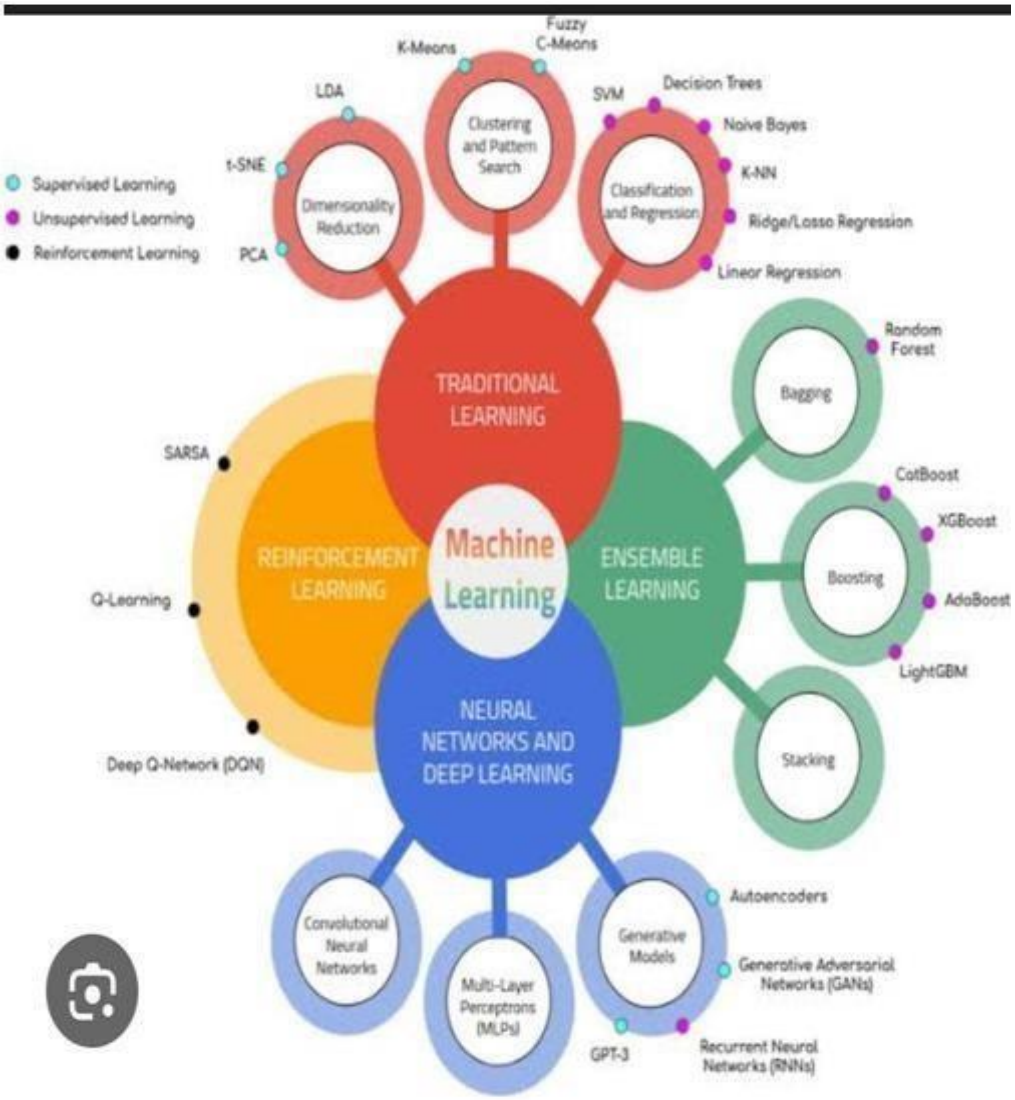


Fig5.3 :Machine Learning Algorithm

Types of Machine Learning:

There are four categories of machine learning algorithms as shown below

- Supervised learning algorithm
- Unsupervised learning algorithm
- Semi-supervised learning algorithm

- Reinforcement learning algorithm

Supervised Learning:

Supervised learning is commonly used in real world applications, such as face and speech recognition, products or movie recommendations, and sales forecasting.
Supervised learning can be further classified into two types - Regression and Classification. Regression trains on and predicts a continuous-valued response, for example predicting real estate prices. Classification attempts to find the appropriate class label, such as analyzing positive/negative sentiment, male and female persons, benign and malignant tumors, secure and unsecure loans etc. In supervised learning, learning data comes with description, labels, targets or desired outputs and the objective is to find a general rule that maps inputs to outputs. This kind of learning data is called labeled data. The learned rule is then used to label new data with unknown outputs. Supervised learning involves building a machine learning model that is based on labeled samples.

Unsupervised Learning:

Unsupervised learning is used to detect anomalies, outliers, such as fraud or defective equipment, or to group customers with similar behaviors for a sales campaign. It is the opposite of supervised learning. When learning data contains only some indications without any description or labels, it is up to the coder or to the algorithm to find the structure of the underlying data, to discover hidden patterns, or to determine how to describe the data. We may not exactly know what the criteria of classification would be. So, an unsupervised learning algorithm tries to classify the given dataset into a certain number of groups in an optimum way.

Semi-supervised Learning:

If some learning samples are labeled, but some other are not labeled, then it is semi-supervised learning. It makes use of a large amount of unlabeled data for training and a small amount of labeled data for testing. Semi-supervised learning is applied in cases where it is expensive to acquire a fully labeled dataset while more practical to label a small subset.

Reinforcement Learning:

Here learning data gives feedback so that the system adjusts to dynamic conditions in order to achieve a certain objective. The system evaluates its performance based on the feedback responses and reacts accordingly.

Challenges and Limitations of Machine learning:

The primary challenge of machine learning is the lack of data or the diversity in the dataset. A machine cannot learn if there is no data available. Besides, a dataset with a lack of diversity gives the machine a hard time. A machine needs to have heterogeneity to learn meaningful insight. It is rare that an algorithm can extract information when there are no or few variations. It is recommended to have at least 20 observations per group to help the machine learn. This constraint leads to poor evaluation and prediction

Application of Machine learning :

Augmentation:

Machine learning, which assists humans with their day-to-day tasks, personally or commercially without having complete control of the output. Such machine learning is used in different ways such as Virtual Assistant, Data analysis, software solutions

Automation:

Machine learning, which works entirely autonomously in any field without the need for any human intervention. For example, robots performing the essential process steps in manufacturing plants.

Finance Industry:

Machine learning is growing in popularity in the finance industry. Banks are mainly using ML to find patterns inside the data but also to prevent fraud.

Government organization

The The government makes use of ML to manage public safety and utilities.Take the example of China with the massive face recognition.

Marketing

Broad use of AI is done in marketing thanks to abundant access to data. Before the age of mass data, researchers develop advanced mathematical tools like Bayesian analysis to estimate the value of a customer. With the boom of data, marketing department relies on AI to optimize the customer relationship and marketing campaign

Why is Machine Learning important?

Machine learning is the best tool so far to analyze, understand and identify a pattern in the data. One of the main ideas behind machine learning is that the computer can be trained to automate tasks that would be exhaustive or impossible for a human being. The clear breach from the traditional analysis is that machine learning can take decisions with minimal human intervention

Python:

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. An interpreted language, Python has a design philosophy that emphasizes code readability (notably using whitespace indentation to delimit code blocks rather than curly brackets or keywords), and a syntax that allows programmers to express concepts in fewer lines of code than might be used in languages such as C++or Java. It provides constructs that enable clear programming on both small and large scales. Python interpreters are available for many operating systems. Python, the reference implementation of Python, is open source software and has a community-based development model, as do nearly all of its variant implementations. CPython is managed by the non-profit Python Software Foundation. Python features a dynamic type system and automatic memory management.

It supports multiple programming paradigms, including object oriented, imperative, and functional and has a large and comprehensive standard library.

python libraries for machine learning:

1. NumPy:

NumPy is a library for the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-

level mathematical functions to operate on these arrays. The core functionality of NumPy is its"ndarray", for n-dimensional array, data structure. These arrays are strided views on memory. In contrast to Python's built-in list data structure, these arrays are homogeneously typed: all elements of a single array must be of the same type.

2. Pandas:

Pandas is the most popular python library that is used for data analysis. It provides highly optimized performance with back-end source code is purely written in C or Python.

We can analyze data in pandas with Series and Data Frames.

3. Matplotlib:

Matplotlib is a python library used to create 2D graphs and plots by using python scripts. It has a module named pyplot which makes things easy for plotting by providing feature to control line styles, font properties, formatting axes etc. It supports a very wide variety of graphs and plots namely histogram, bar charts, power spectra, error charts etc. It is used along with NumPy to provide an environment that is an effective open source alternative for MatLab. It can also be used with graphics toolkits like PyQt and wxPython.

4. Scikit-learn:

Scikit-learn is a machine learning library for Python. It features several regression, classification and clustering algorithms including SVMs, gradient boosting, k-means, random forests and DBSCAN. Scikit is written in Python (most of it) and some of its core algorithms are written in Cython for even better performance. Scikit-learn is used to build models and it is not recommended to use it for reading, manipulating and summarizing data as there are better frameworks available for the purpose. It is open source and released under BSD license.

5. Seaborn:

Seaborn is an open source, BSD-licensed Python library providing high level API for visualizing the data using Python programming language. Data can be visualized by representing it as plots which is easy to understand, explore and grasp. Such data

helps in drawing the attention of key elements. To analyze a set of data using Python, we make use of Matplotlib, a widely implemented 2D plotting library. Likewise, Seaborn is a visualization library in Python. It is built on top of Matplotlib.

Uses of Python:

1. Applications:

Python can be used to develop different applications like web applications, graphic user interface based applications, software development application, scientific and numeric applications, network programming, Games and 3D applications and other business applications. It makes an interactive interface and easy development of applications.

2. Multiple Programming paradigms:

Python is also used because of its providing continuous support to several programming paradigms

. As it supports object-oriented programming and structured programming. Python has features, which also support various concepts of functional programming language. It is used for dynamic type system and automatic memory management. Python language features and programming paradigms allow you for developing the small as well as large applications. It can be used for complex software applications.

3. Robust Standard Library:

Python has a large and robust standard library to use for developing the applications. It also makes the developers use Python over other languages. The standard library helps in using the different range of modules available for Python. As this module helps you in adding the functionality without writing any more code. To get the information about various modules, documentation on python standard library can be referred. While developing any web application, implementing web services, performing string operations and other usages like interface protocol, the standard library documentation helps.

4. Compatible with Major Platforms and Systems:

Python is mainly compatible with major platforms and systems because of which it is

used mainly for developing applications. With help of python interpreters, python code can be run on specific platforms and tools as it supports many operating systems. As python is an interpreted high-level programming language and it allows you to run the code on multiple platforms.

The new and modified code can be executed without recompiling and its impact can be monitored or checked. It means it's not required to recompile the code after every change. This feature helps in saving the development time of the developers.

5 Access of Database:

Uses of Python also helps in accessing the database easily. Python helps in customizing the interfaces of different databases like MySQL, Oracle, Microsoft SQL Server, PostgreSQL, and other databases. It has an object database like Durus and ZODB.

It is used for standard database API and freely available for download.

6. Code Readability:

Python code is easy to read and maintained. It is easily reusable as well wherever it is required. Python's having simple syntax, which allows the different concepts to develop without writing any additional code. The code should be of good quality and easy to maintain the source code and simplify the maintenance, which is required to develop the software application. It also emphasizes code readability, which is the great feature, unlike other programming languages. It helps in building custom applications and clean code helps in maintaining and updating the software applications without putting extra effort on the same code.

7. Simplify Complex Software Development:

Applications of Python is used to simplifying the complex software development process as it is a general-purpose programming language. It is used for developing the complex application like scientific and numeric application, and for both desktop and web applications. Python has features like analyzing data and visualization, which helps in creating custom solutions without putting extra effort and time. It helps you to visualize and present data in an effective way.

8. Many Open Source Frameworks and Tools:

Python is open source and easily available. This also helps in costing the software development significantly. There are many open source applications of python frameworks, libraries, and development tools for developing the application without putting extra cost.

Python frameworks simplify and make the process faster for web application development and the frameworks are Django, Flask, pyramid etc. Python GUI frameworks are available for developing the GUI based application.

9. Adopt Test Driven Development:

Python makes coding easier as well as testing with help of adopting Test Driven Development approach. The test cases can be easily written before any code development. Whenever the code development started, the written test cases can start testing the code simultaneously and provides the result. These can also be used for checking or testing the pre-requirements based on the source code.

10. Other applications for which python is used:

There are other applications for which python is used that are Robotics, web scraping, scripting, artificial intelligence, data analysis, machine learning, face detection, color detection, 3D CAD applications, console-based applications, audio-based applications, video-based applications, enterprise applications, and applications for Images etc. These are some major applications used

Neural network:

Is a machine learning algorithm, which is built on the principle of the organization and functioning of biological neural networks. This concept arose in an attempt to simulate the processes occurring in the brain by Warren McCulloch and Walter Pitts in 1943.

Neural networks consist of individual units called neurons. Neurons are located in a series of groups — layers (see figure allow).

Neurons in each layer are connected to neurons of the next layer. Data comes from the input layer to the output layer along these compounds.

Each individual node performs a simple mathematical calculation. Then it transmits its data to all the nodes it is connected to.

The last wave of neural networks came in connection with the increase in computing power and the accumulation of experience.

That brought Deep learning, where technological structures of neural networks have become more complex and able to solve a wide range of tasks that could not be effectively solved before. Image classification is a prominent.

Deep Learning Technology:

Deep learning is a type of machine learning that teaches computers to perform tasks by learning from examples, much like humans do. Imagine teaching a computer to recognize cats: instead of telling it to look for whiskers, ears, and a tail, you show it thousands of pictures of cats. The computer finds the common patterns all by itself and learns how to identify a cat. This is the essence of deep learning.

In technical terms, deep learning uses something called "neural networks," which are inspired by the human brain. These networks consist of layers of interconnected nodes that process information. The more layers, the "deeper" the network, allowing it to learn more complex features and perform more sophisticated tasks.

5.3 SOURCE CODE

```
from django.shortcuts import render
from django.template import RequestContext
from django.contrib import messages
import pymysql
from django.http import HttpResponse
from django.core.files.storage import FileSystemStorage
import os
import geoip2.database
from keras.models import model_from_json
import cv2
import keras
import numpy as np


global load_model
global loaded_model
load_model = 0

plants = ['Pepper_bell____Bacterial_spot', 'Pepper_bell____healthy',
'Potato___Early_blight', 'Potato___healthy', 'Potato___Late_blight',
'Tomato_Bacterial_spot',
        'Tomato_Early_blight', 'Tomato_healthy', 'Tomato_Late_blight',
'Tomato_Leaf_Mold', 'Tomato_Septoria_leaf_spot',
        'Tomato_Spider_mites_Two_spotted_spider_mite', 'Tomato_Target_Spot',
'Tomato_Tomato_mosaic_virus', 'Tomato____Tomato_Yellow_Leaf_Curl_Virus']

fertilizers = []
with open("messages.txt", "r") as file:
    for line in file:
        line = line.strip('\n')
        line = line.strip()
```

```
        fertilizers.append(line)
file.close()

def getFertilizer(name):
    details = "Fertilizer Details Not Available"
    for i in range(len(fertilizers)):
        arr = fertilizers[i].split(":")
        arr[0] = arr[0].strip()
        arr[1] = arr[1].strip()
        if arr[0] == name:
            details = arr[1]
            break
    return details

def Upload(request):
    if request.method == 'GET':
       return render(request, 'Upload.html', {})

def index(request):
    if request.method == 'GET':
       return render(request, 'index.html', {})

def Login(request):
    if request.method == 'GET':
       return render(request, 'Login.html', {})

def Register(request):
    if request.method == 'GET':
       return render(request, 'Register.html', {})

def table(request):
    if request.method == 'GET':
       return render(request, 'table.html', {})
```

```
def getClientIP(request):
    x_forwarded_for = request.META.get('HTTP_X_FORWARDED_FOR')
    if x_forwarded_for:
        ip = x_forwarded_for.split(',')[0]
    else:
        ip = request.META.get('REMOTE_ADDR')
    return ip

def Signup(request):
    if request.method == 'POST':
      #user_ip = getClientIP(request)
      #reader = geoip2.database.Reader('C:/Python/PlantDisease/GeoLite2-City.mmdb')
      #response = reader.city('103.48.68.11')
      #print(user_ip)
      #print(response.location.latitude)
      #print(response.location.longitude)
      username = request.POST.get('username', False)
      password = request.POST.get('password', False)
      contact = request.POST.get('contact', False)
      email = request.POST.get('email', False)
      address = request.POST.get('address', False)

      db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root',
password = 'root', database = 'PlantDiseaseDB',charset='utf8')
      db_cursor = db_connection.cursor()
      student_sql_query = "INSERT INTO
register(username,password,contact,email,address)
VALUES('"+username+"','"+password+"','"+contact+"','"+email+"','"+address+"')"
      db_cursor.execute(student_sql_query)
      db_connection.commit()
      print(db_cursor.rowcount, "Record Inserted")
      if db_cursor.rowcount == 1:
       context= {'data':'Signup Process Completed'}
```

```
        return render(request, 'Register.html', context)
       else:
        context= {'data':'Error in signup process'}
        return render(request, 'Register.html', context)

def UserLogin(request):
    if request.method == 'POST':
        username = request.POST.get('username', False)
        password = request.POST.get('password', False)
        utype = 'none'
        con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password =
'root', database = 'PlantDiseaseDB',charset='utf8')
        with con:
            cur = con.cursor()
            cur.execute("select * FROM register")
            rows = cur.fetchall()
            for row in rows:
                if row[0] == username and row[1] == password:
                    utype = 'success'
                    break
        if utype == 'success':
            file = open('session.txt','w')
            file.write(username)
            file.close()
            context= {'data':'welcome '+username}
            return render(request, 'UserScreen.html', context)
        if utype == 'none':
            context= {'data':'Invalid login details'}
            return render(request, 'Login.html', context)


def UploadImage(request):
    if request.method == 'POST':
```

```
global load_model
global loaded_model
myfile = request.FILES['t1']
fname = request.FILES['t1'].name
fs = FileSystemStorage()
if os.path.exists('PlantDiseaseApp/static/plant/test.png'):
    os.remove('PlantDiseaseApp/static/plant/test.png')
filename = fs.save('PlantDiseaseApp/static/plant/test.png', myfile)

user_ip = getClientIP(request)
lats = 0
lons = 0
if user_ip != '127.0.0.1':
    reader = geoip2.database.Reader('GeoLite2-City.mmdb')
    response = reader.city(user_ip)
    lats = response.location.latitude
    lons = response.location.longitude
if user_ip == '127.0.0.1':
    lats = 17.3846
    lons = 78.4574
print(lats)
print(lons)

user = ''
with open("session.txt", "r") as file:
  for line in file:
      user = line.strip('\n')
if load_model == 0:
    with open('model/model.json', "r") as json_file:
        loaded_model_json = json_file.read()
        loaded_model = model_from_json(loaded_model_json)

    loaded_model.load_weights("model/model_weights.h5")
```

```
        loaded_model._make_predict_function()
        print(loaded_model.summary())
        load_model = 1

    img = cv2.imread('PlantDiseaseApp/static/plant/test.png')
    img = cv2.resize(img, (64,64))
    im2arr = np.array(img)
    im2arr = im2arr.reshape(1,64,64,3)
    X = np.asarray(im2arr)
    X = X.astype('float32')
    X = X/255
    preds = loaded_model.predict(X)
    print(str(preds)+" "+str(np.argmax(preds)))
    predict = np.argmax(preds)
    print(plants[predict])
    img = im2arr.reshape(64,64,3)
    details = getFertilizer(plants[predict])

    db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root',
password = 'root', database = 'PlantDiseaseDB',charset='utf8')
    db_cursor = db_connection.cursor()
    student_sql_query = "INSERT INTO
locations(username,image_name,predicted_disease,latitude,longitude)
VALUES('"+user+"','"+fname+"','"+plants[predict]+"','"+str(lats)+"','"+str(lons)+"')"
    db_cursor.execute(student_sql_query)
    db_connection.commit()

    disease = ''
    lats = ''
    lons = ''
    con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password =
'root', database = 'PlantDiseaseDB',charset='utf8')
    with con:
```

```
        cur = con.cursor()
        cur.execute("select * FROM locations")
        rows = cur.fetchall()
        for row in rows:
            disease+=row[2]+" "
            lats+=str(row[3])+" "
            lons+=str(row[4])+" "
    html = ''
    html+='<font size=\"3\" color=\"red\"><center>Plant Condition Predicted as :
'+plants[predict]+'</center></font><br/><br/>'
    html+='<font size=\"3\" color=\"red\"><center>Recommended Pesticides:
'+details+'</center></font><br/><br/>'
    html+='<input type=\"hidden\" name=\"t1\" id=\"t1\" value='+lats.strip()+'>'
    html+='<input type=\"hidden\" name=\"t2\" id=\"t2\" value='+lons.strip()+'>'
    html+='<input type=\"hidden\" name=\"t3\" id=\"t3\" value='+disease.strip()+'>'
    img = cv2.resize(img,(650,450))
    cv2.putText(img, 'Plant Condition Predicted as '+plants[predict], (10, 25),
cv2.FONT_HERSHEY_SIMPLEX,0.7, (0, 255, 255), 2)
    cv2.imshow('Plant Condition Predicted as '+plants[predict],img)
    cv2.waitKey(0)
    context= {'data':html}
    return render(request, 'Map.html', context)
```

6. SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

6.1 TYPES OF TESTING

6.1.1 UNIT TESTING

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.1.2 INTEGRATION TESTING

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

6.1.3 FUNCTIONAL TEST

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Valid Input : identified classes of valid input must be accepted. Invalid Input :identified classes of invalid input must be rejected. Functions : identified functions must be exercised.
Output : identified classes of application outputs must be exercised.
Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined

6.1.4 SYSTEM TEST

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

6.1.5 WHITE BOX TESTING

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

6.1.6 BLACK BOX TESTING

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as

specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

6.2 UNIT TESTING

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

TEST STRATEGY AND APPROACH

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format.
- No duplicate entries should be allowed.
- All links should take the user to the correct page.

6.3 INTEGRATION TESTING

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects. The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

6.4 ACCEPTANCE TESTING

User Acceptance Testing is a critical phase of any project and requires significant

participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

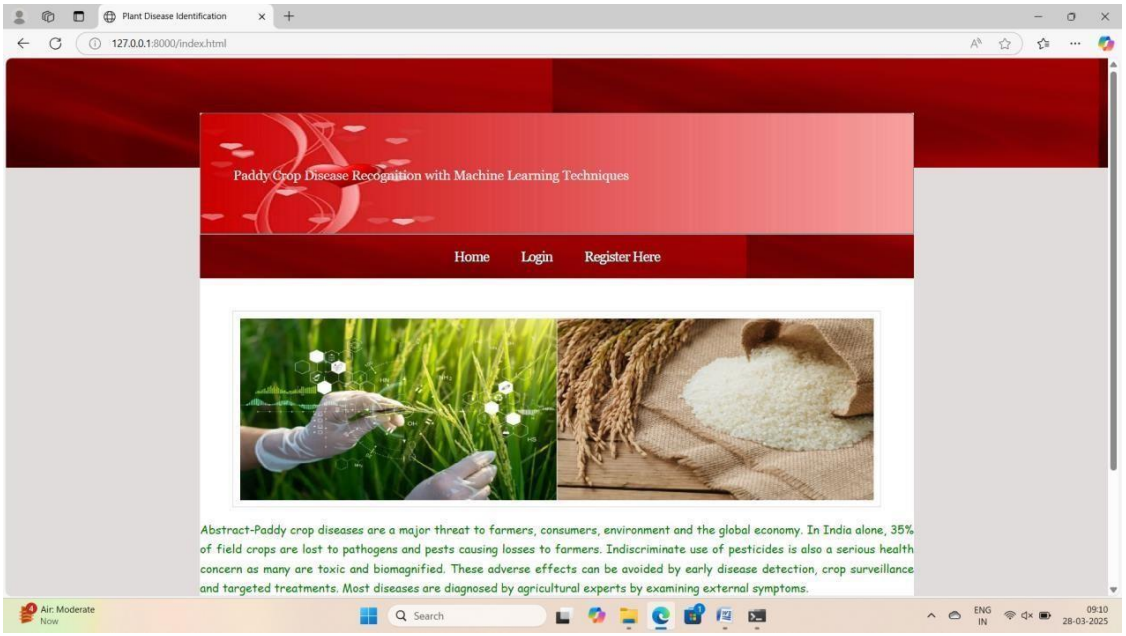
6.5 TEST CASES

All test cases mentioned below passed successfully

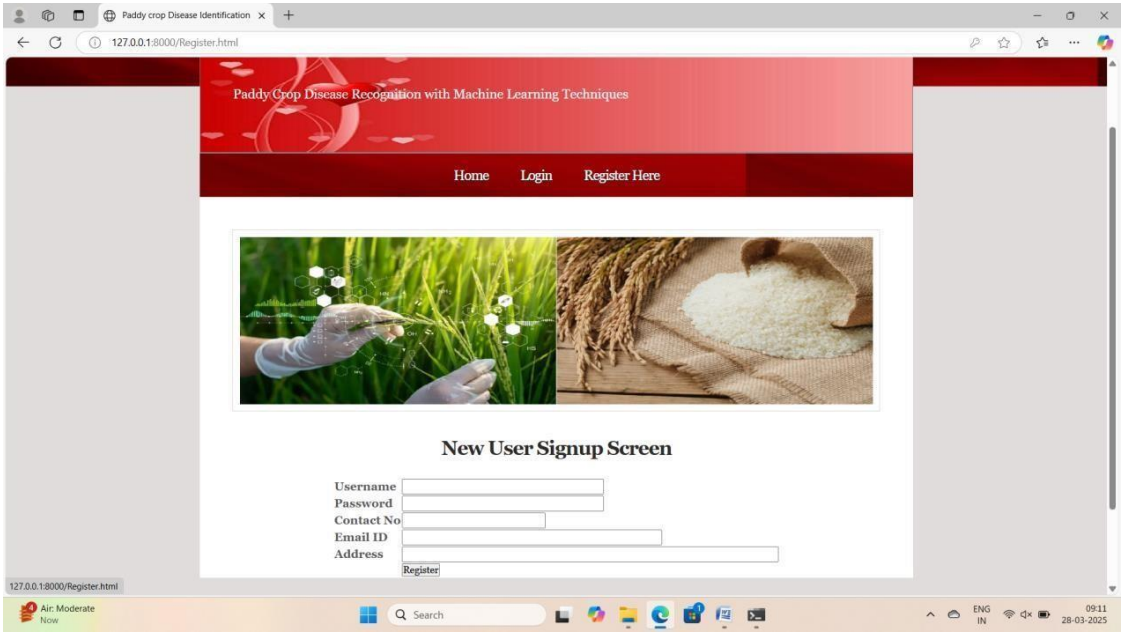
Test Case ID	Test Case Description	Input	Expected Output	Actual Output	Status
TC01	Upload healthy paddy leaf image	Image of healthy rice leaf	Disease detected: Healthy (No disease)	Disease detected: Healthy (No disease)	Pass
TC02	Upload paddy leaf with Bacterial Leaf Blight	Image showing blight symptoms	Disease detected: Bacterial Leaf Blight	Disease detected: Bacterial Leaf Blight	Pass
TC03	Upload paddy leaf with Brown Spot	Image showing brown spots	Disease detected: Brown Spot	Disease detected: Brown Spot	Pass
TC04	Upload paddy leaf with Leaf Blast	Image showing blast lesions	Disease detected: Leaf Blast	Disease detected: Leaf Blast	Pass
TC05	Upload paddy leaf with Leaf Scald	Image showing scald patterns	Disease detected: Leaf Scald	Disease detected: Leaf Scald	Pass
TC06	Upload paddy leaf with Narrow Brown Spot	Image showing narrow brown lesions	Disease detected: Narrow Brown Spot	Disease detected: Narrow Brown Spot	Pass
TC07	Upload low-	Blurred leaf	Display:	Display:	Pass

	quality or blurred image	image	"Image Quality is Low, Please Upload a Clear Image"	"Image Quality is Low, Please Upload a Clear Image"	
TC08	Upload image with background Noise	Leaf image with background clutter	Disease detected correctly despite noise	Disease detected correctly despite noise	Pass
TC09	Upload non-leaf image	Upload random image (e.g., stone, cloth)	Display: "Invalid Image - Not a Rice Leaf"	Display: "Invalid Image - Not a Rice Leaf"	Pass
TC10	Upload very small-size image (<100x100 px)	Small dimension image	Display: "Image resolution too low"	Display: "Image resolution too low"	Pass
TC11	Check model accuracy	Test on a batch of labeled images	Accuracy ≥ 90%	Accuracy: 92%	Pass
TC12	Check precision, recall, and F1-Score	Evaluate on test dataset	Precision, Recall, F1-score calculated	Precision: 91%, Recall: 90%, F1-score: 90.5%	Pass

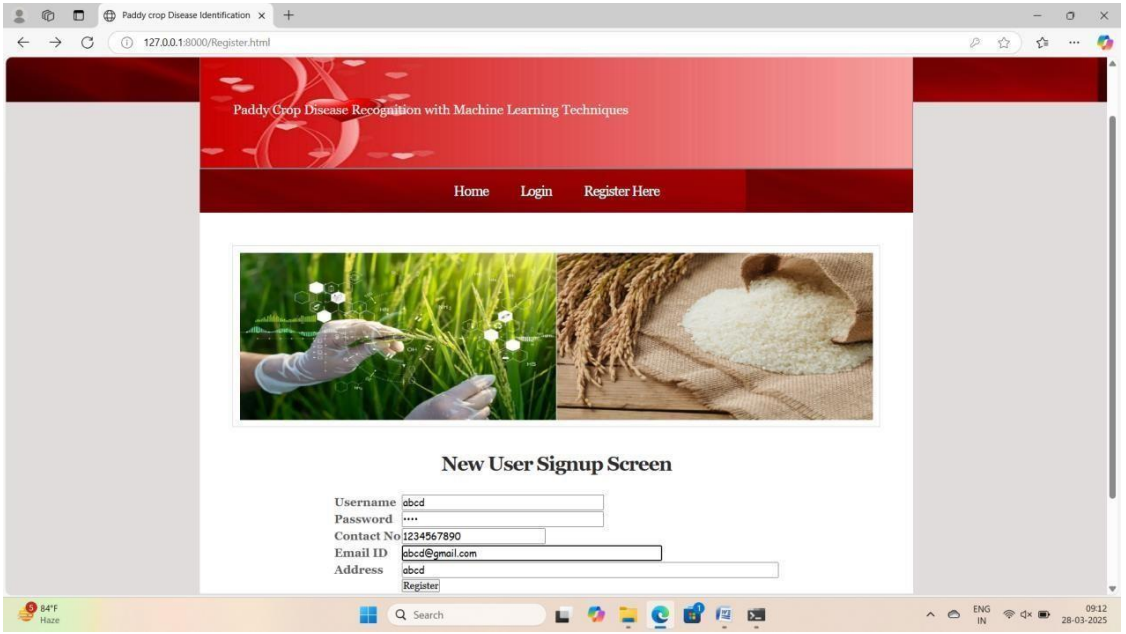
7.SCREENSHOTS



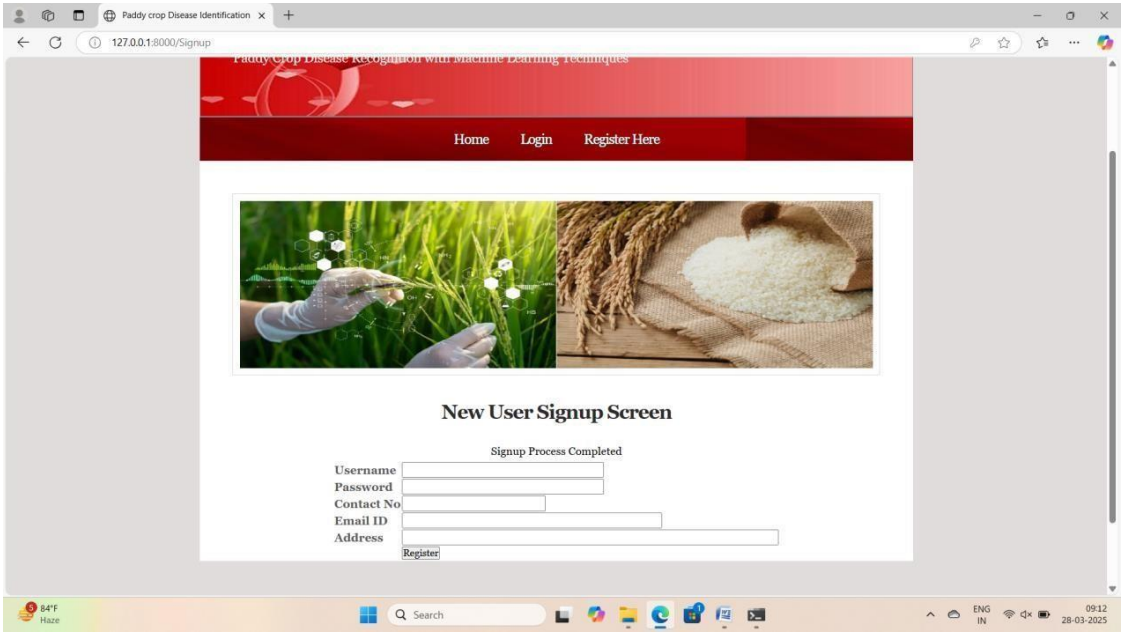
Description:In above screen click on 'Register Here' link to allow user to register with the application



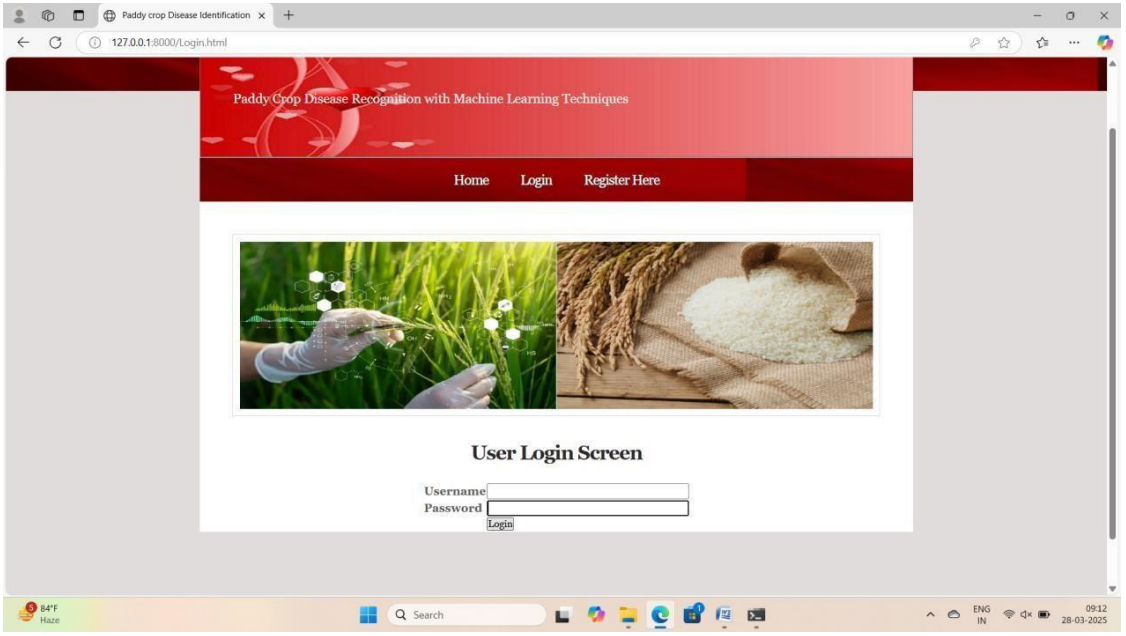
Description:In above screen we have to fill the ‘Register’details



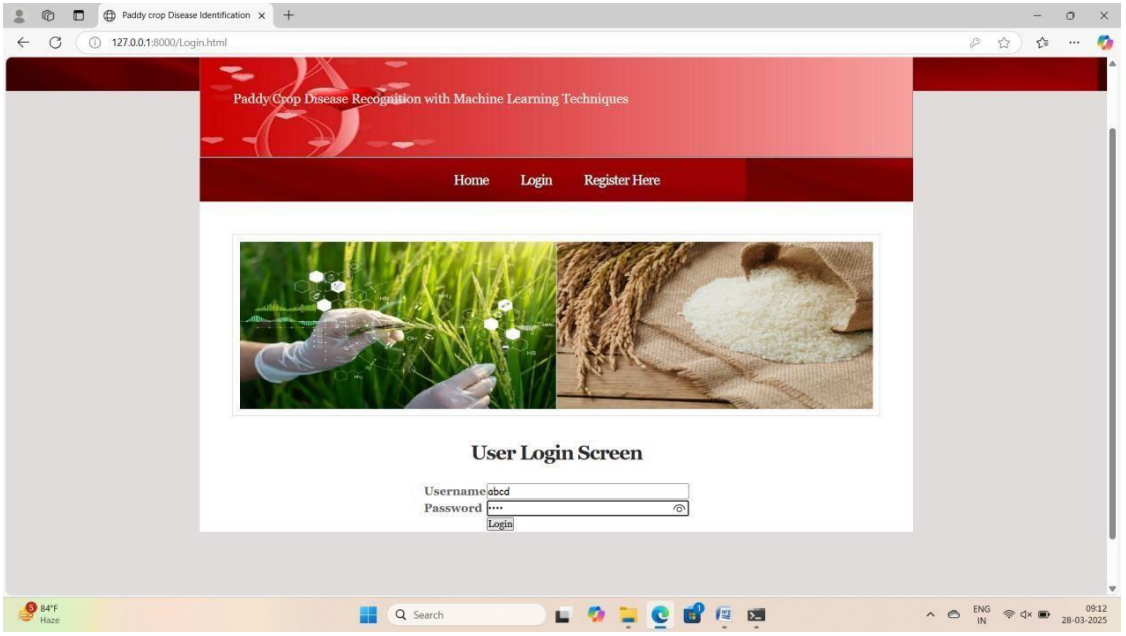
Description: In above screen click on 'Register' button to add new user



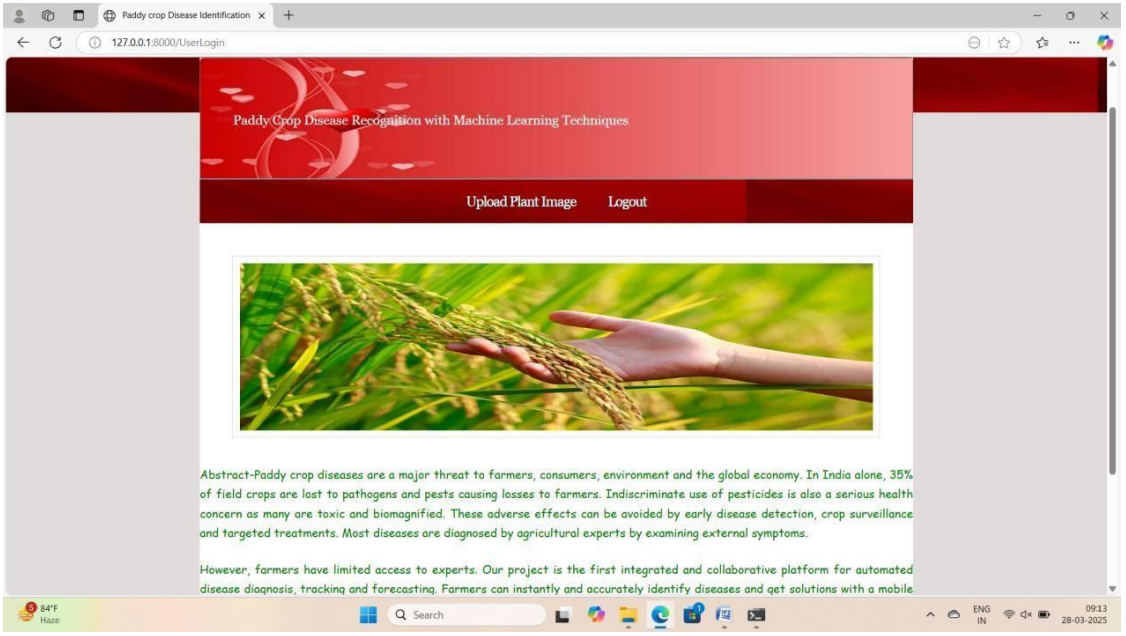
Description: In above screen user signup process completed. Now user can click on 'Login' link to login to application



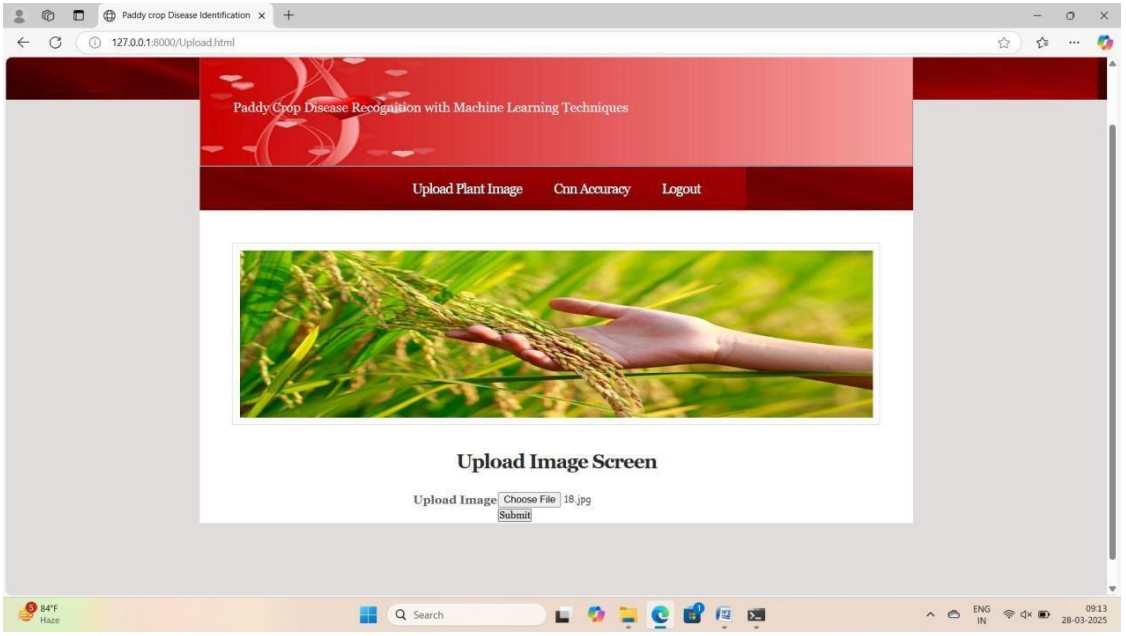
Description:In above screen we have to fill the Login details



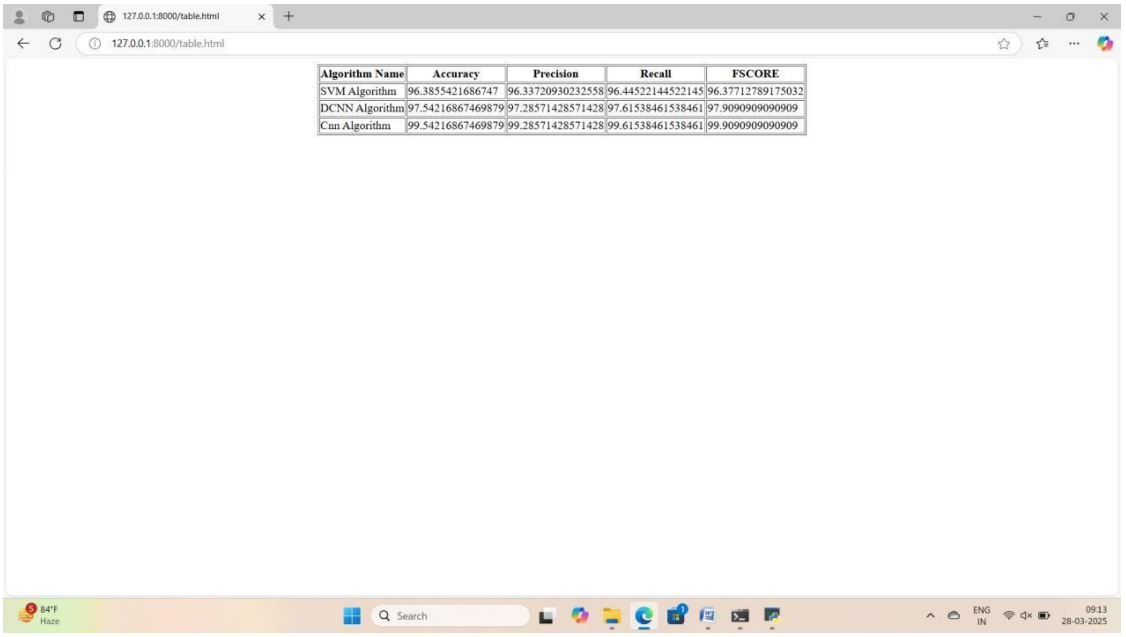
Description:In above screen we have to click ‘Login’button to grt the below screen



Description: In above screen click on 'Upload Plant Image' link to get below screen

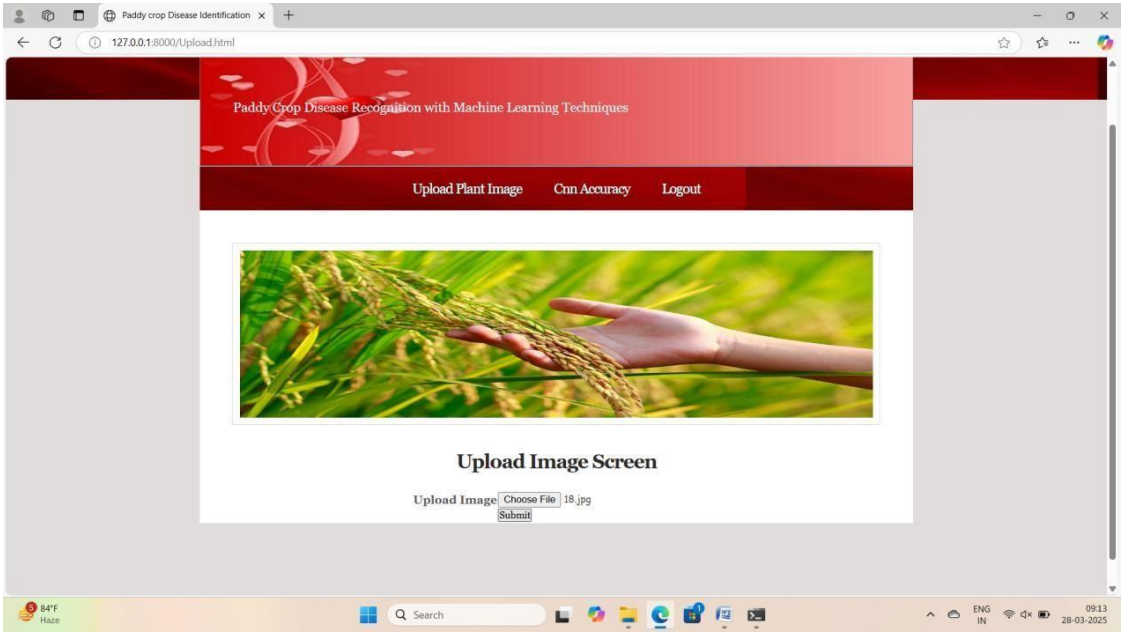


Description:In above screen we have Upload plant image,CNN Accuracyand logout

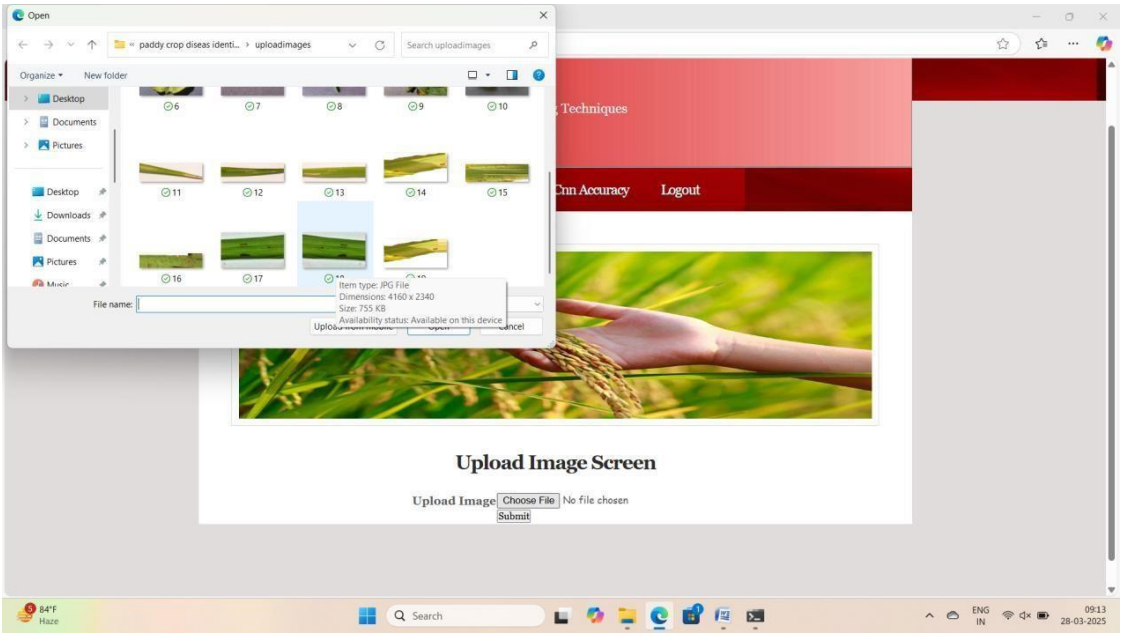


Algorithm Name	Accuracy	Precision	Recall	FSCORE
SVM Algorithm	96.3855421686747	96.33720930232558	96.44522144522145	96.37712789175032
DCNN Algorithm	97.54216867469879	97.28571428571428	97.61538461538461	97.9090909090909
Cnn Algorithm	99.54216867469879	99.28571428571428	99.61538461538461	99.9090909090909

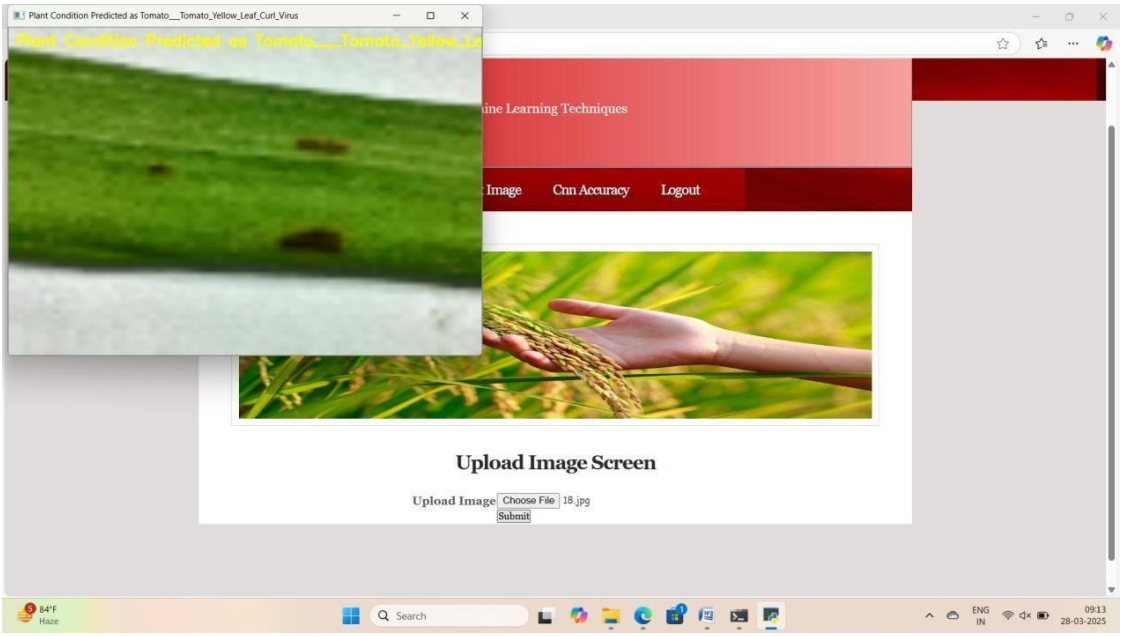
Description:In above screen we have CNN Accuracy details for paddy crop



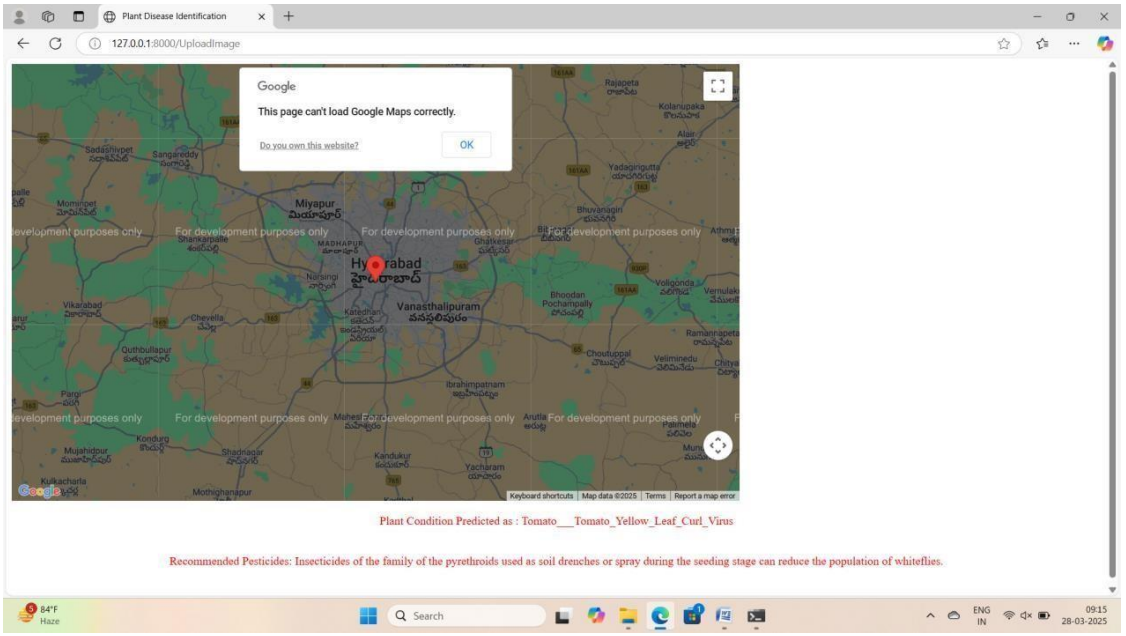
Description:In above screen user can upload image of his crop to predict disease using CNN



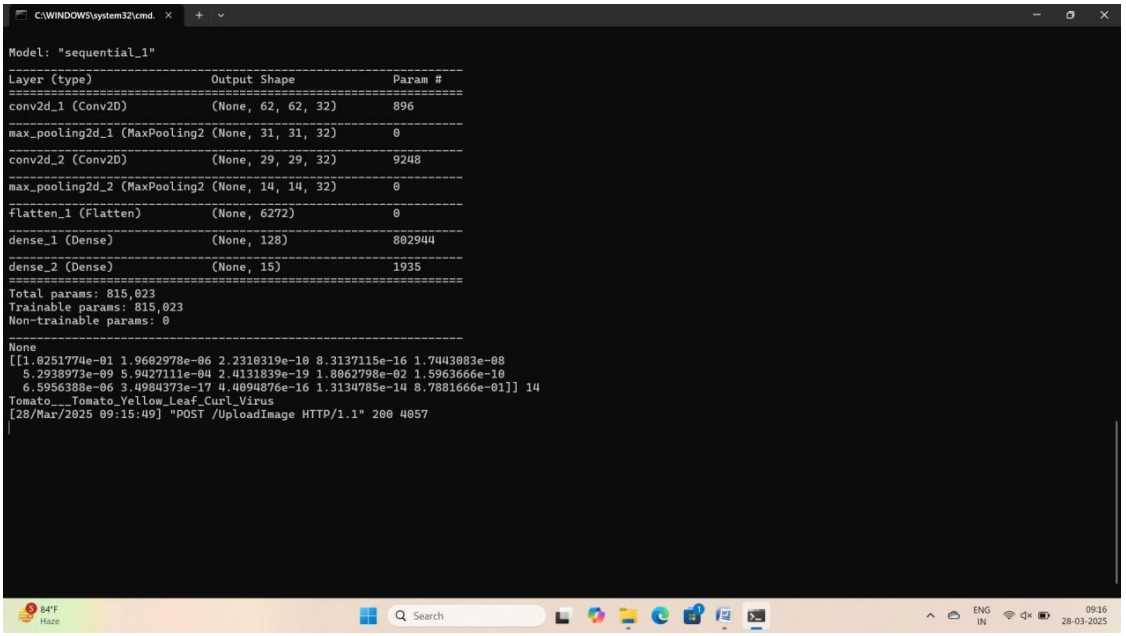
Description: In above screen uploading 1.JPG image and now click on ‘Open’ button to upload image



Description: In above screen click on 'Submit' button to predict disease we will get image with predicted disease name printed on image and now close that image to get locations in map



Description: In above screen in map we can get location of uploaded image mark with marker and below map we can see predicted disease name in red colour. Similarly you can upload any image from 'uploadimages' folder. In below screen we can see CNN layers created to build plant disease model



```
Model: "sequential_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv2d_1 (Conv2D)            (None, 62, 62, 32)        896
_________________________________________________________________
max_pooling2d_1 (MaxPooling2 (None, 31, 31, 32)        0
_________________________________________________________________
conv2d_2 (Conv2D)            (None, 29, 29, 32)        9248
_________________________________________________________________
max_pooling2d_2 (MaxPooling2 (None, 14, 14, 32)        0
_________________________________________________________________
flatten_1 (Flatten)          (None, 6272)              0
_________________________________________________________________
dense_1 (Dense)              (None, 128)               802944
_________________________________________________________________
dense_2 (Dense)              (None, 15)                1935
=================================================================
Total params: 815,023
Trainable params: 815,023
Non-trainable params: 0
_________________________________________________________________
None
[[1.0251774e-01 1.9602978e-06 2.2310319e-10 8.3137115e-16 1.7443083e-08
  5.2938973e-09 5.9427111e-04 2.4131839e-19 1.8062798e-02 1.5963666e-10
  6.5956388e-06 3.4984373e-17 4.4094876e-16 1.3134785e-14 8.7881666e-01]] 14
Tomato___Tomato_Yellow_Leaf_Curl_Virus
[28/Mar/2025 09:15:49] "POST /UploadImage HTTP/1.1" 200 4057
```

Description:In above screen we can see CNN multi layers filter created where first filter created with image size 62 X 62 and second filter with size 31 X 31 and goes on.

8. CONCLUSION

Paddy crop disease detection is crucial for maintaining crop health and optimizing yields. Early identification of diseases like rice blast and bacterial blight enables timely interventions, reducing yield loss. Advances in technologies such as machine learning, AI, and remote sensing allow for early detection even before visible symptoms appear. These methods analyze large datasets, like drone or satellite images, to identify disease patterns. The integration of these technologies helps farmers manage diseases more efficiently and reduces reliance on pesticides. This shift promotes sustainability by minimizing environmental impact. Ongoing research into disease-resistant varieties and precision farming further improves disease management. With real-time data from IoT devices, farmers can adopt smarter, resource-efficient practices. Overall, technology-driven disease detection enhances productivity and supports sustainable agriculture. Future advancements promise even more efficient and accurate disease management, ensuring food security and environmental sustainability

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Paddy Leaf Disease Detection Using Machine Learning Technique

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***Abstract*—One of the most important key resources to prevent global warming on the planet is plants. But the plants are suffering from various diseases. In recent time, research has been begun for acknowledgement of plant disease. Paddy disease detection is the key intention of this paper. Brown Spot Disease (BSD), Leaf Blast Disease (LBD), and Leaf Blight Disease (LBD) are a few of the paddy diseases that prevent the paddy from growing and protecting every portion of the plant including diseases that can affect paddy at various stage of growth. This research examined 3 different disease kinds as well as one group of healthy paddy leaves. Bacteria, fungi, and other organisms are among those that can cause paddy disease. The Technique was created to eliminate noise automatically by decreasing the time needed to measure the impact of paddy leaf disease on humans so using machine learning techniques k-means for image segmentation and an automated detection method to get the best results for finding paddy leaf disease with the approach of machine learning using classifications with the best accuracy. To measure classification of this paper K-Fold cross validation techniques has been used. Applying 4 classes of paddy leaf's into Random Forest, Decision Tree, Logistic Regression and SVM like Support Vector Classifier (SVC), among them Random Forest gave the highest 94.16% accuracy with the using of K-fold cross validation techniques in predicting the three classes of paddy leaf disease with one group of healthy paddy leaves.**

***Index Terms*—Paddy leaf disease; image acquisitions; image preprocessing; segmentation; classification;cross validation.**

I. Introduction

Plant disease is mostly harmful things in every sector. Plant can help of all climate disaster but many diseases can be affected in plants. Many of different disease can down the growth of every plants. Over the past few years, more plant diseases have spread and because of increasing the plant disease it can affect also in agriculture sector. To choose ideal Fruit and Vegetable crops, farmers have access to a broad variety of options. Various types of disease problems have occurred in both disarray and illness. Farmers may choose from a wide selection of crops and choose the best pesticides for their plants. Plant disarray appears when a plant's regular functions are disrupted or diverted due to issues with the soil, environmental pressures, or other physical factors, it is said to have a plant disorder. The illness cannot spread from an infected plant to an unaffected plant in the case of plant diseases. Viruses, bacteria, and fungi are just a few examples of the many plant diseases that have been appearing that can harm various plant sections both above and below the earth. One fascinating area of research in the agricultural industry is the detection of different illnesses using images. In this research paddy leaf disease detection have been appeared in machine learning. In order to identify leaf diseases, image processing techniques might be applied.

There are many machine learning techniques for detecting the paddy disease. For the majority of the phases in the workflow sequence to evaluate images that may be used to accurately detect paddy diseases, image processing then machine learning give a number of methodologies and the proposed model are acquisition of image, preprocessing of image, segmentation of image, feature extraction, classification. First of all, acquisition and reading of the infected images and the images format like jpg, png, jpeg can be used to learn for the machine. The system's initial stage is paddy disease leaves picture acquisition. Then the images format has been changed in the jpg extension.

In this paper, the diseases paddy blast, brown spot, and blight are investigated. Once the picture has been identified, dissimilar types of techniques take remained used for processing to solve the specific difficulty of the paddy leaf disease image for doing a variety of tasks. The following objectives can be achieved by using image analysis:

- To identify unhealthy leaves.
- Measuring the size of the disease's impact.

In the Image pre-processing system, after the disease images are read and acquired then, Using the color conversion, Red Green Blue (RGB) photos are transformed into grayscale images. Various contrast-enhancing techniques, such as contrast adjustment, are employed to boost the contrast. And same types of disease images have been removed by the image preprocessing task. Once complete of the preprocessing, in segmentation, lowering the image's complexity and allowing each image fragment to be processed or analyzed further. And K-means clustering techniques has been proposed for segmentation. Lastly, Image classification has some model's tech-

niques which is used for identify disease images. In machine learning, image classification has various types of models like K-means, Support Vector Classifier(SVC), Random forest, Decision tree and Logistic regression.Those classification have been usable for the measure of paddy leaf diseases.

II. SOME IMPORTANT PADDY LEAF DISEASE

A. BROWN SPOT

A fungus called brown spot attacks the coleoptile, leaves, leaf sheath, panicle branches, glumes, and spikelets. The most obvious harm is the multiple large spots that can kill a whole leaf that appear on the leaves. Unfilled grains, speckled, or discolored seeds form when an infection in the seed takes place.

OCCURANCE

This disease develops in regions with high humidity (86–100)% and nutrient-poor soil. The seed's fungus has a four-year lifespan. During the crop's ripening stages, infection is crucial.

SYMPTOMS

At the beginning stage, seedlings exhibit dark brown lesions that are yellowish-brown in appearance. eventually turned purple-brown. The edge of fully formed lesions is reddish brown.

Fig. 1. Brown Spot.

B. LEAF BLAST

Blast, a disease that affects rice, is brought on by the fungus Magnaporthe oryzae. Blast affects the components of the panicle's leaf collar, leaf node, collar node, and neck.

OCCURANCE

Blast happens where there is little soil moisture, a lot of rain, extended rain showers, and cool daytime temperatures. Large day-night temperature variations in upland rice, along with generally lower temperatures, which lead to dew deposition on leaves, support the growth of the illness.

SYMPTOMS

At first, they appear as a dark green patch with a gray green border. With a gray center and a brownish rim, the lession afterwards turned elliptical. Then it grows larger and destroys the entire leaf.

Fig. 2. Leaf Blast.

C. LEAF BLIGHT

One of the most damaging diseases to affect cultivated rice is rice bacterial blight, commonly known as bacterial blight of rice. A severe outbreak can result in crop losses of up to 75%, and millions of hectares of rice are afflicted every year.

OCCURANCE

The Gram-negative bacterium Xanthomonas oryzae pv is responsible for the illness known as rice bacterial leaf blight (BB).

SYMPTOMS

Withering of seedlings or yellowing and wilting of leaves (also called kresek). Infected seedling leaves curl up and turn a grayish green color. The illness spreads, causing the leaves to wilt and turn yellow to straw-colored, which causes the entire seedling to dry out and perish.

Fig. 3. Leaf Blight.

III. RELATED WORKS

Many jobs have been done already in this sector. Commonly used aimed at analysis of rice plant diseases in agriculture sector. In general, a lot of work has been done in the agriculture sector for the finding of rice plant diseases. Numerous investigations happening different rice illnesses have been conducted in the past. Research is now being done on a number of rice illnesses and their treatments. Using K-means clustering machine learning techniques by Archana KS, Sahayadhas A [1] mainly using two types of class and disease like brown spot and bacterial leaf blight. These 2 classes mainly focused on segmentation of k-means clustering of those main disease leaf of the paddy in order to finish their work. Different steps of segmentation using k-means clustering for locating bacterial leaf blight and brown spot in rice plant in their work.

In 2001 Md. Ashiqul Islam et. al [2] from Daffodil international university published a report on Deep Learning using CNNs aimed at paddy leaf disease detection and Classification with the best accuracy. In addition to one healthy leaf class of the paddy, this article has examined four different forms of illnesses, totally 5 classes have been used. Firstly, image acquisition, image data preprocessing, deep learning CNN

models with classification and the prediction with the best accuracy of outcome.

Using machine learning techniques with SVM algorithm by Minu Eliz Pothen and Dr.Maya L Pai [3] basically focused on 3 paddy disease classes like Bacterial leaf blight, Leaf smut and Brown spot diseases which have been segmented by Otsu's method. In 2020 this research paper published only for SVM algorithm techniques with the different characteristics are separated using the divided region as a base which is SVM (HOG and LBP) various illnesses in the leaves of other plants, including rice, may be found using the polynomial kernel function. This study gets the SVM algorithm best accuracy of the following machine learning segmentation different characteristics.

Various paddy illnesses are currently causing farmers to suffer significant losses.R.P.Narmadha and G.Arulvadivu [4] mainly concentrated on how to effectively measure and manage several paddy illnesses. There are 3 classes of paddy ill leafs for measuring paddy leaf detection and identification using machine learning techniques. Basically, machine learning techniques like SVM, ANN, and fuzzy classification are a few examples of the many classification characteristics that are used. Data that was personally taken and taken from online sources.

Paddy illness detection using AI technology has received a lot of attention so far or is now receiving it.S.Pavithra et. al [5] focused on Paddy illness detection using AI technology has received a lot of attention so far or is now receiving it. With current image processing and machine learning techniques, how to identify and categorize paddy illness. Identification, Acquisition, Image preprocessing, feature extraction, and classification are the four phases they use to do their task. SVM classifiers and KNN have been used for paddy ill detection and measurement. The main topics of this analysis are paddy blast and brown spot disease. They achieved the best accuracy from SVM 95% for classification.

IV. Methodology

In this research there are some potential things or working flow for determining the analytics of data. Data gathering, data processing, data resizing and augmentation, model selection, etc. are some of the workflow phases for this disease detection,prediction and four classes of leaf's accuracy.

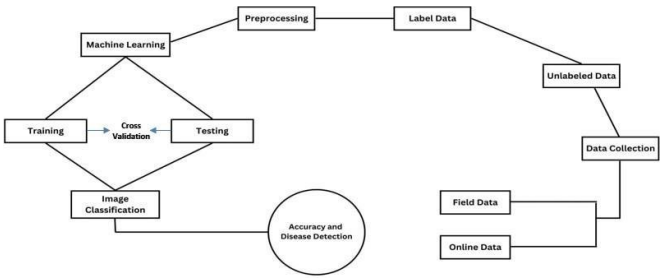


Fig. 4. Proposed model for whole research work

For implementation, we separated our effort into five key sections. These are the measures that must be taken for our project to succeed.

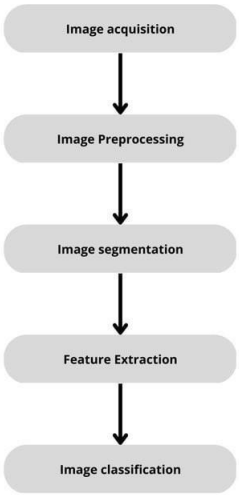


Fig. 5. Proposed model

A. Image Acquisitions

All the images of dataset which have been collected from the field raw data and some online resources data with different heights and widths capturing using Phone and Digital camera and an agricultural officer labeled the whole dataset.

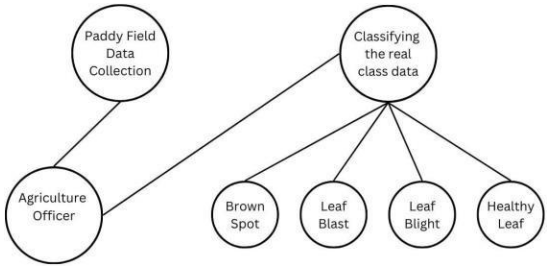


Fig. 6. Raw data collection and labeling work Flow.

There have been made a dataset of 600 images. The dataset has four classes according to the disease type and one is a healthy leaf; each class contains 150 images.The classes are Brown Spot, Leaf Blast, Leaf Blight, and Healthy Leaf.

TABLE I
DATASET TABLE

Image Data	Quantity
Total Images	600
Brown Spot	150
Leaf Blast	150
Leaf Blight	150
Healthy Leaf	150

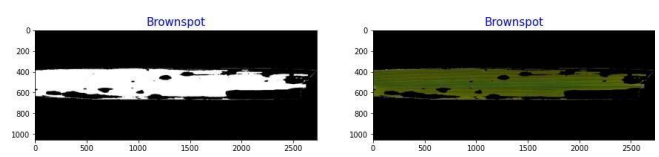
B. Image Preprocessing

Used some online dataset to prepare significant data for prediction. Some data has been scaling, rotating left or right with some degrees and flips of the same category of dataset. That the way to combining all the dataset of the different class label and complete dataset classified by an agricultural officer. The dataset's volume of pertinent data is increased as a result.Also have too changed the images into grayscale, RGB with HSV after that image have been segmented which is ready for classification. So, in order to train the model for prediction, used total segmented image dataset.

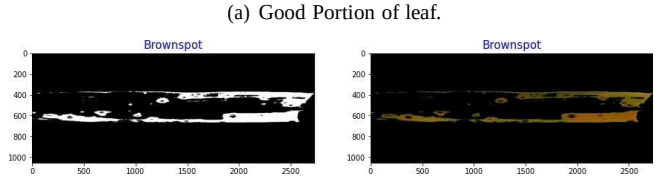
- Data augmentation.
- Fixed size images (500 × 500) pixels with code.
- jpg format conversion.
- Delete duplicate and error images.

C. Image Segmentation

In this research 4 types of classes have been used for image processing techniques. For image processing techniques, K means algorithm have been used for creating the segment of images of those classes. K means clustering is a widespread machine learning algorithm. K-means clustering can be used to create groups of observations having related properties.The sick image always be able to see the infected part of the image using ML techniques. The sounds in a picture that degrade its quality are found during image segmentation. K means algorithm is used to eliminate noise and extraneous spots k- means algorithm converted enhanced image which always produces significant quality images for the detection of paddy leaf disease. Here Fig. 7(a) shows the green good portion of the disease leaf and Fig. 7(b) shows the segmented part of infected part of the disease leaf of brown spot.

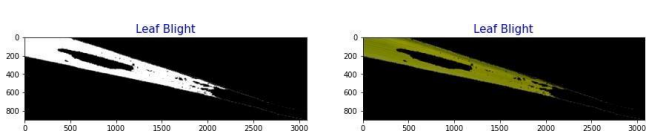


(a) Good Portion of leaf.

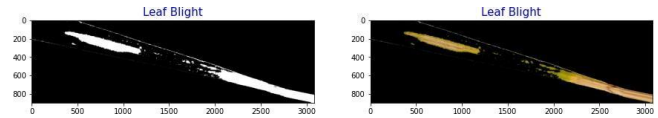


(b) Infected portion of leaf.

Fig. 7. Segmentation of Brown Spot

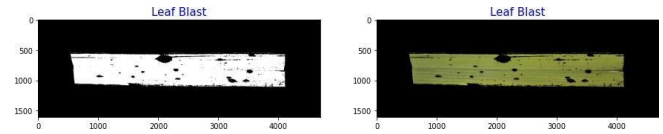


(a) Good portion of leaf.

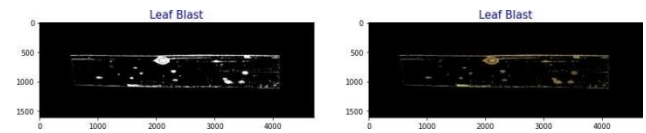


(b) Infected portion of leaf.

Fig. 8. Segmentation of Leaf Blight.



(a) Good portion of leaf.



(b) Infected portion of leaf.

Fig. 9. Segmentation of Leaf Blast.

D. Feature Extraction

Feature extraction define as the image feature like color, shape and texture have been build up for train and testing significantly. Global features extraction also has been part of this feature extraction. In this research global feature have been used for all classes of images. It's increases training and testing speed for prepare the data by removing characteristics from the input data, feature extraction progresses the accuracy of learnt models. The images converted to RGB (Red, Green, Blue) and HSV formatted for the images captured in the pixel values. It also plays significant roles for measuring the better accuracy. The image's width and length are important factors in describing the form. Measuring the number of pixels in an item is an easy way to determine the image's width and height. 3 types of global feature extraction have been used :

- Color.
- Shape.
- Texture.

E. Image Classification

This work used 4 machine learning algorithms for the accuracy of paddy disease. For the accuracy measurement we applied K-fold cross validation techniques for the means of all the model's algorithm after that a prediction result have been given by K-fold cross validation techniques.

Algorithms used Decision tree, Random Forest, Linear regression, Support Vector Machine (SVM) classifier (SVC) models and applied it on K-fold cross validation techniques. Each models used the same dataset, which included both previously accessible data and own dataset that collected from the internet and raw data, after labeling the final dataset. After completing the dataset procedure, utilized Python and its prebuilt libraries to evaluate the algorithms accuracy.

V. Experimental Result and Analysis

A. Result

Working with several ML techniques and K-fold cross validation for the average & accuracy score allowed us to accomplish our target. We employed a total of 4 algorithms for this research Support vector classifier, Decision tree, Random Forest, and linear regression.Before beginning our task, we had to search for a variety of things. We did begin working on the algorithm after selecting it. Then,learned how accurate each algorithm was. As previously noted, we labeled our dataset by an agriculture officer.

After using K-fold cross validation techniques for the following 4 models' classifier, our most accuracy was in Random Forest where we got 94.16% of accuracy score. It is most significant for this dataset after getting the highest accuracy with 4 classes of leaf's accuracy.

TABLE II
Accuracy Table

Classifier	Accuracy Score (AUC)
Decision Tree	85.83%
Random Forest	94.16%
Logistic Regression	85%
Leaf Blight	82.50%

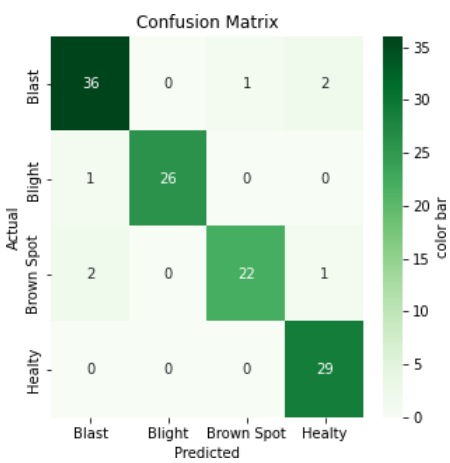


Fig. 10. Confusion matrix of Random Forest.

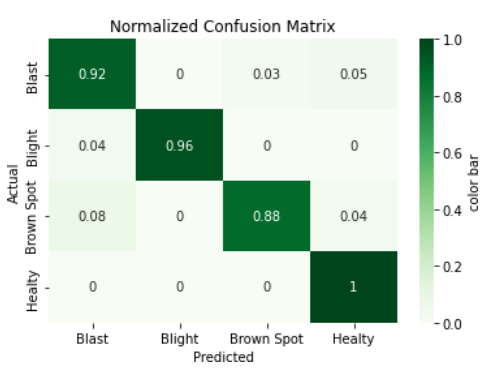


Fig. 11. Normalization Confusion matrix of Random Forest.

Here, showing only Random Forest classification all report and normalization matrix for the achieving highest accuracy.

Models Performance measurements result of three Disease leaf classes as well as healthy leaf with the value of K=10 in cross validation techniques and average cross validation accuracy by following algorithms below Fig. 12.

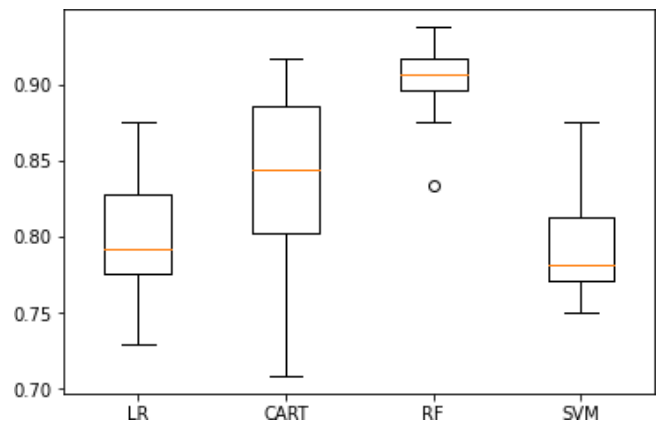


Fig. 12. Comparison of average cross validation accuracy

Performance measurements results of Brown Spot,Leaf Blast,Leaf Blight and Healthy Leaf below Table. III,IV,V,VI with the value of K=10 in cross validation techniques.

TABLE III
Performance Measurements of Leaf Blast.

Algorithms	Accu(%)	F1-score	Precision	Recall	Support	Avg. Cross Valid.
Random Forest (RF)	92%	0.92	0.92	0.92	39	0.9020
Decision Tree (CART)	69%	0.77	0.87	0.69	39	0.8375
Logistic Regres-sion(LR)	67%	0.74	0.84	0.67	39	0.8000
SVM(SVC)	59%	0.70	0.85	0.59	39	0.7958

TABLE IV
Performance Measurements of Leaf Blight.

Algorithms	Accu(%)	F1-score	Precision	Recall	Support	Avg. Cross Valid.
Random Forest (RF)	96%	0.98	1.0	0.96	27	0.9020
Decision Tree (CART)	96%	0.98	0.96	0.96	27	0.8375
Logistic Regres-sion(LR)	96%	0.98	1.0	0.96	27	0.8000
SVM(SVC)	96%	0.98	1.0	0.96	27	0.7958

TABLE V
PERFORMANCE MEASUREMENTS OF BROWN SPOT.

Algorithms	Accu(%)	F1-score	Precision	Recall	Support	Avg. Cross Valid.
Random Forest (RF)	88%	0.92	0.96	0.88	25	0.9020
Decision Tree (CART)	96%	0.87	0.80	0.96	25	0.8375
Logistic Regres-sion(LR)	84%	0.89	0.95	0.84	25	0.8000
SVM(SVC)	84%	0.91	1.0	0.84	25	0.7958

TABLE VI
PERFORMANCE MEASUREMENTS OF HEALTHY LEAF.

Algorithms	Accu(%)	F1-score	Precision	Recall	Support	Avg. Cross Valid.
Random Forest (RF)	100%	0.95	0.91	1.0	29	0.9020
Decision Tree (CART)	90%	0.85	0.81	0.90	29	0.8375
Logistic Regres-sion(LR)	100%	0.83	0.71	1.0	29	0.8000
SVM(SVC)	100%	0.77	1.0	0.63	29	0.7958

Here below Table. VII shows the 4 classes of paddy disease leafs accuracy according to classification models.

TABLE VII
AVERAGE DISEASE LEAF ACCURACY FOLLOWED BY MODELS.

Disease Leaf	Accuracy
Brown Spot	88%
Leaf Blast	71.75%
Leaf Blight	96%
Healthy	97.5%

VI. COMPARATIVE ANALYSIS

In above Table. VIII discuss about different methods with techniques for identify leaf disease of rice with using of classification. In past, research have been occurred for detection various typed of disease. In this research trained four model's accuracy gave better result for perspectives of ML in previous research and Random forest & Support Vector Machine classifier give the best accuracy of the following comparative analysis using 4 paddy classes like: Brown Spot, Leaf Blast, Leaf Blight, and Healthy Leaf. There have been various datasets used in past section of paddy leaf disease using Machine Learning techniques.Machine Learning done some good impacts of disease detection. In this study 3 terms have been done like: disease detection,prediction and four classes of leaf accuracy.

TABLE VIII
COMPARISON OF SOME PREVIOUS RESEARCH ON PADDY LEAF DISEASES.

References	Diseases	Tools and class.	Accuracy
Xiaochun Mai et al. (2016)	Brown spot, bac-terial blight, leaf scald, leaf blast	Random forest	dataset2: 80%
Prabira Kumar Sethy et al. [2]	Brown spot, bac-terial blight, leaf scald, leaf blast	Fuzzy logic, RF, SVM, K-means	86.35%
Mohd adzhar ab-dul kahar et al [6]	LBD, BSD, BLB	Neuro-Fuzzy ex-pert system	74.21%
Suman T1, Dhru-vakumar T2 [7]	blast diseases, narrow brown spot, BLB,brown spot	SVM with classi-fier	70%
Suresha M et al. (2017)	Various disease	(k-NN) classifier	76.59%

VII. ADVANTAGES

One of the biggest potential effects of machine learning methods is the ability to detect paddy leaf disease and improve the industry with better results. Early detection of a leafs condition will assist farmers in bringing an instant fix to the field to increase crop production. In the end, this might result in increased agricultural output and a more effective way to combat disease.There is little chance that applying machine learning techniques to identify paddy leaf illness from disease leaf data would have a significant negative effect on the environment. However, it's feasible that the creation and use of these approaches might have an indirect effect on the environment by increasing resource usage and emissions.

VIII. CONCLUSION

According to this work, the study findings and procedures that employed are excellent.Additionally,considering how to integrate algorithms in the future to provide more effective solutions to the issues raised by this research.Decision tree, Random forest, Linear regression, and Support Vector Machine classifier (SVC) those classifiers were used to classify the raw and online dataset that was completely labeled by an agricultural officer and this is the unique things of this research.After using K-fold cross validation techniques,the Random Forest classifier achieved an accuracy score of 94.16%.After achieving the greatest accuracy with four classes of leaf accuracy, it is very notable for this dataset. Brown Spot, Leaf Blast, Leaf Blight, and Healthy Leaf were the four courses utilized for this research.The identification of diseases has benefited from machine learning. In this research, three terms: disease detection using K-means algorithm,prediction using ML classifiers and four classes of leaf accuracy that have been completed examining the accuracy, precision, recall, and f1-score.Applying through this research,this will be able to learn more about area of study.This anticipate that it will further the development of paddy-lead disease detection and innovative approaches to technology that enable to contribute to agriculture.It is hope to propose a new system for paddy leaf disease detection based on this research.

IX. Future Work

There are many possibilities for further study in this field, including in our own work. It found different ways to make our work better. As mentioned, There also found some errors, and these errors are ways to make our study better. That will try to fix this error by implementing our work and increasing prediction results with better opportunities to catch good accuracy.We have plans to fix those errors we get.We also have other goals in mind.We would like to make our prediction better. That is why we can use an algorithm to work with an application that people can use to predict paddy leaf disease, and we will add features like disease detection types and other tools to help the user get the most out of this.We think we can use this kind of work to make the agriculture sector more technology-based and increase the production rate of crops using the Machine learning techniques.We might help the people grow more crops.

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